Deep Learning

IV B.Tech - I Sem (PVP-23)

Learn from Data. Predict the Future.

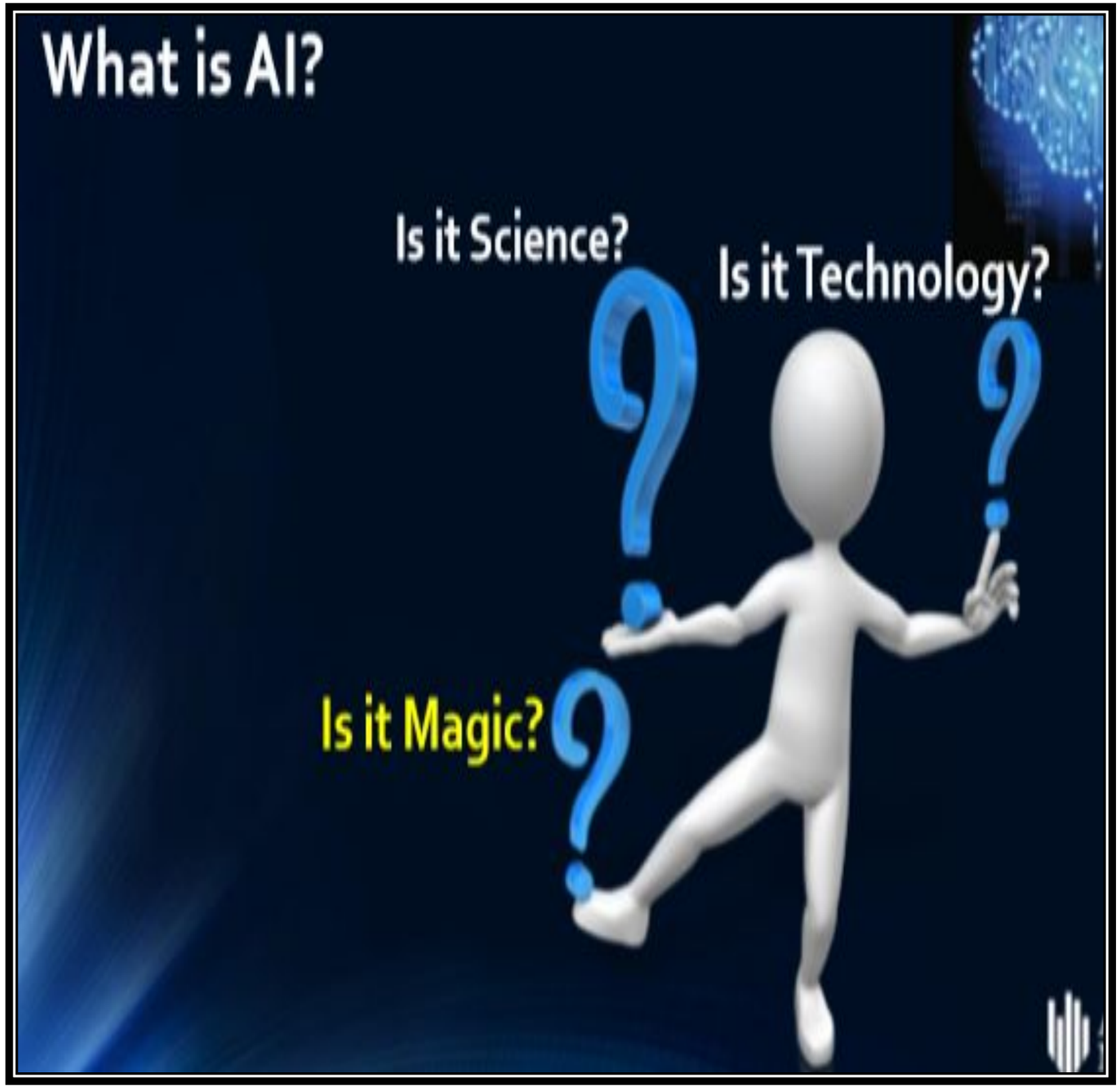


Topics :

- Introduction.
- Why we study the Artificial Intelligence
- Diff B/W Human Intelligence and AI
- Differences b/w AI, ML and DL
- Why we use Deep Learning?
- Key Points in Deep Learning
- Reasons to use Deep Learning
- How Deep Learning Works?
- Example of Deep Learning.
- Applications of Deep Learning

Topics :

- Deep Learning Frame Works
- Learning beyond Curriculum



Artificial Intelligence is a **Science** that uses **technology** to create **its magic**

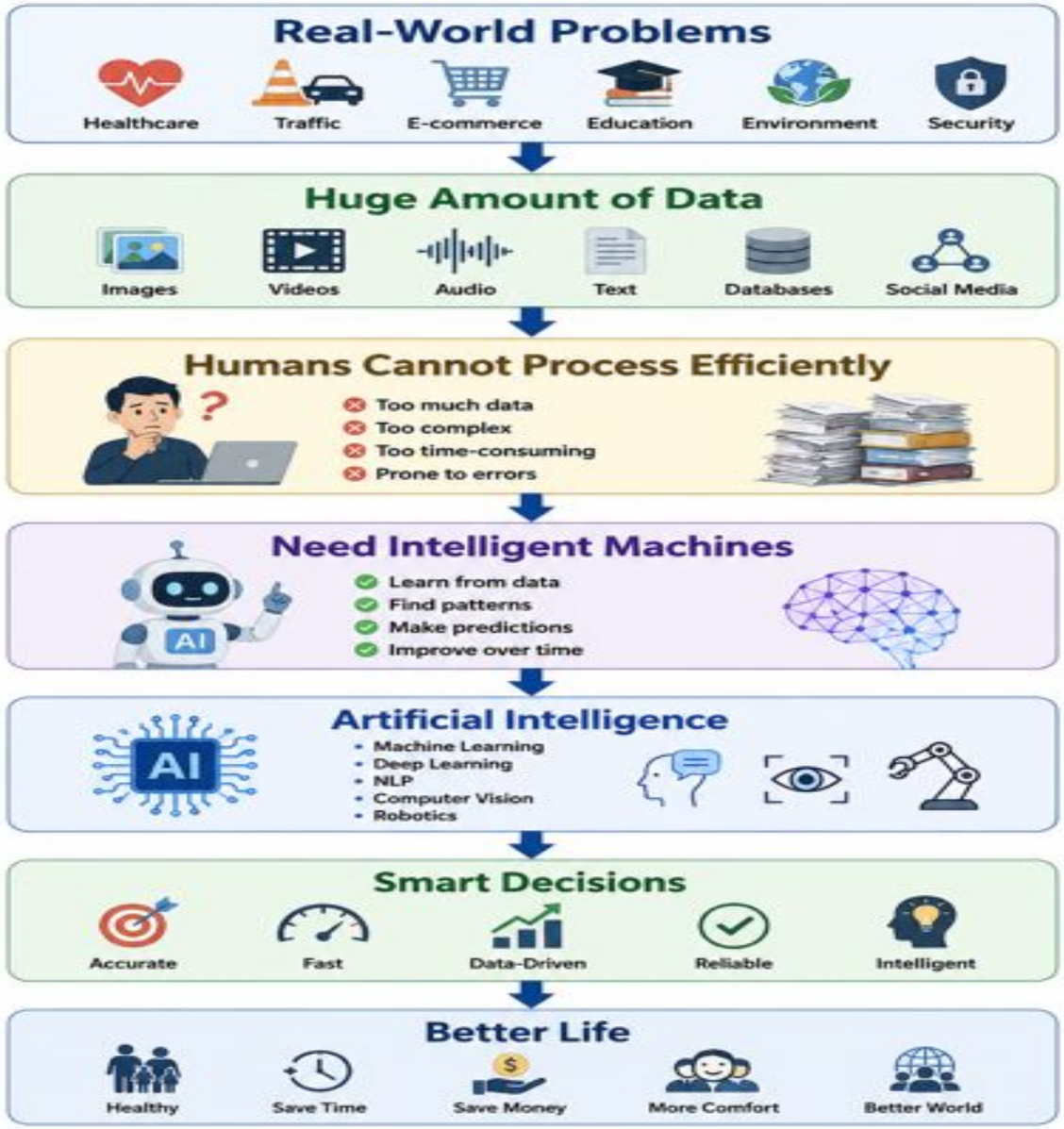
ML **DL**

Definition of AI

✔ Artificial Intelligence is composed of **two words Artificial and Intelligence.**

- Where Artificial defines **”man-made”** and
- intelligence defines **”thinking power”,**
- hence AI means **”a man-made thinking power.”**

Why we need AI



Definition of AI

Artificial Intelligence is needed to **enable machines to perform tasks** that **normally require human intelligence**, such as **learning, reasoning, decision-making, problem-solving**, and **understanding complex data.**

Example

Classroom Example

Teacher: "Can you recognize your friend in a class of 60 students?"

Students: "Yes."

Teacher: "Can you recognize your friend among **10 million photos** in one second?"

Students: "No."

Teacher: "AI can do that. That's why we need Artificial Intelligence."

Differences Between Human Intelligence And Artificial Intelligence

1. Faster Decisions
2. Reduction in Human Error
3. Available in 24X7
4. Speed
5. Memory









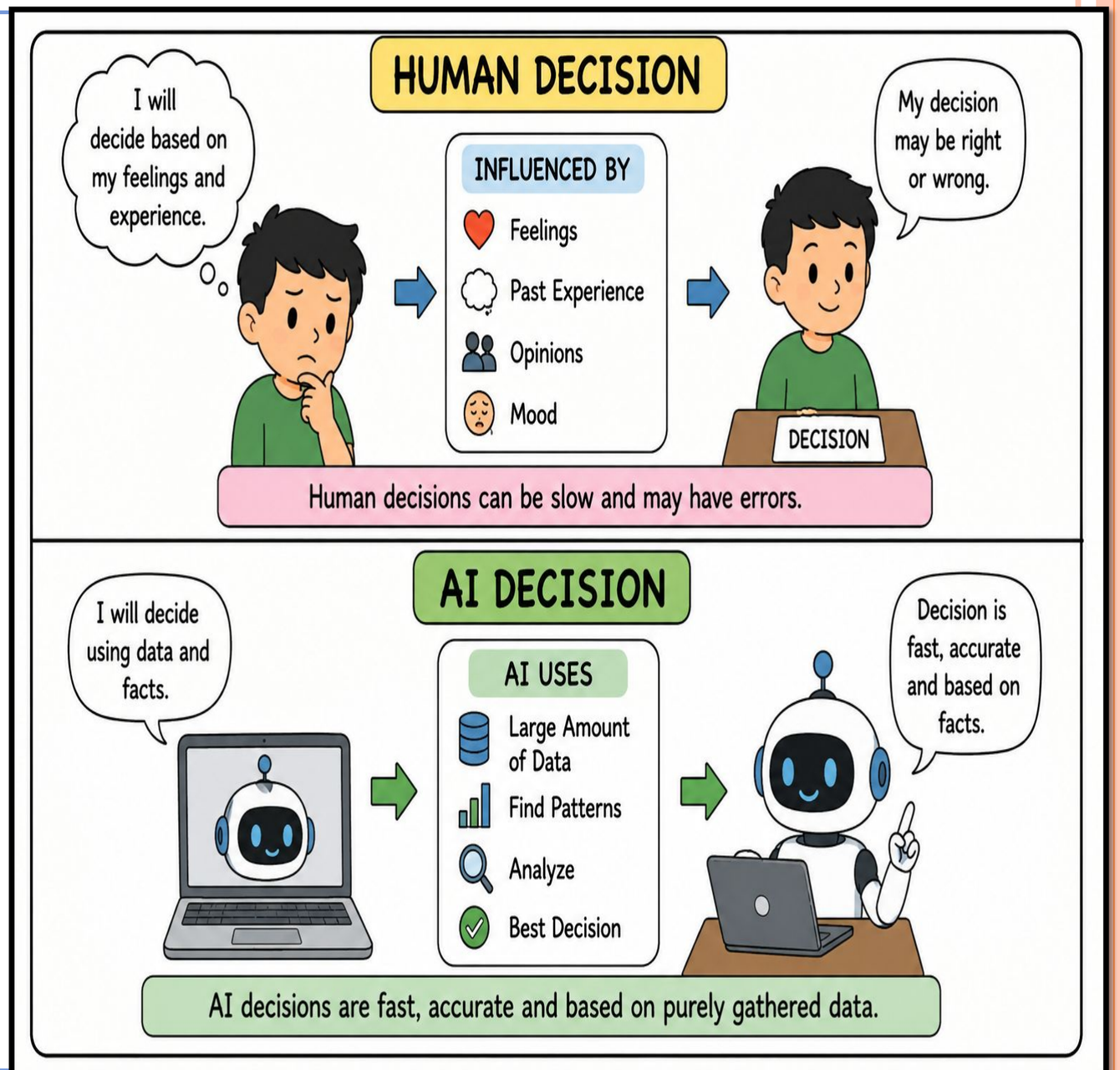




CONTD..

1. **Faster Decisions:** AI is highly objective in decision making as it analyses based on **purely gathered data**. However, humans' decisions may be influenced by subjective elements which are not based on figures alone. human beings are created with the innate **ability to think, reason, recall**, etc.

Example : Recommendation System.



CONTD..

2. Reduction in Human Error : AI often produces accurate results as it functions based on a **set of programmed rules.** As for human intelligence, there is usually a room for **"human error"** as certain details may be missed at one point or the other.

3. Available in 24 x 7: AI can work **24 x 7** , but Humans can work **6 to 8 Hours per day**.

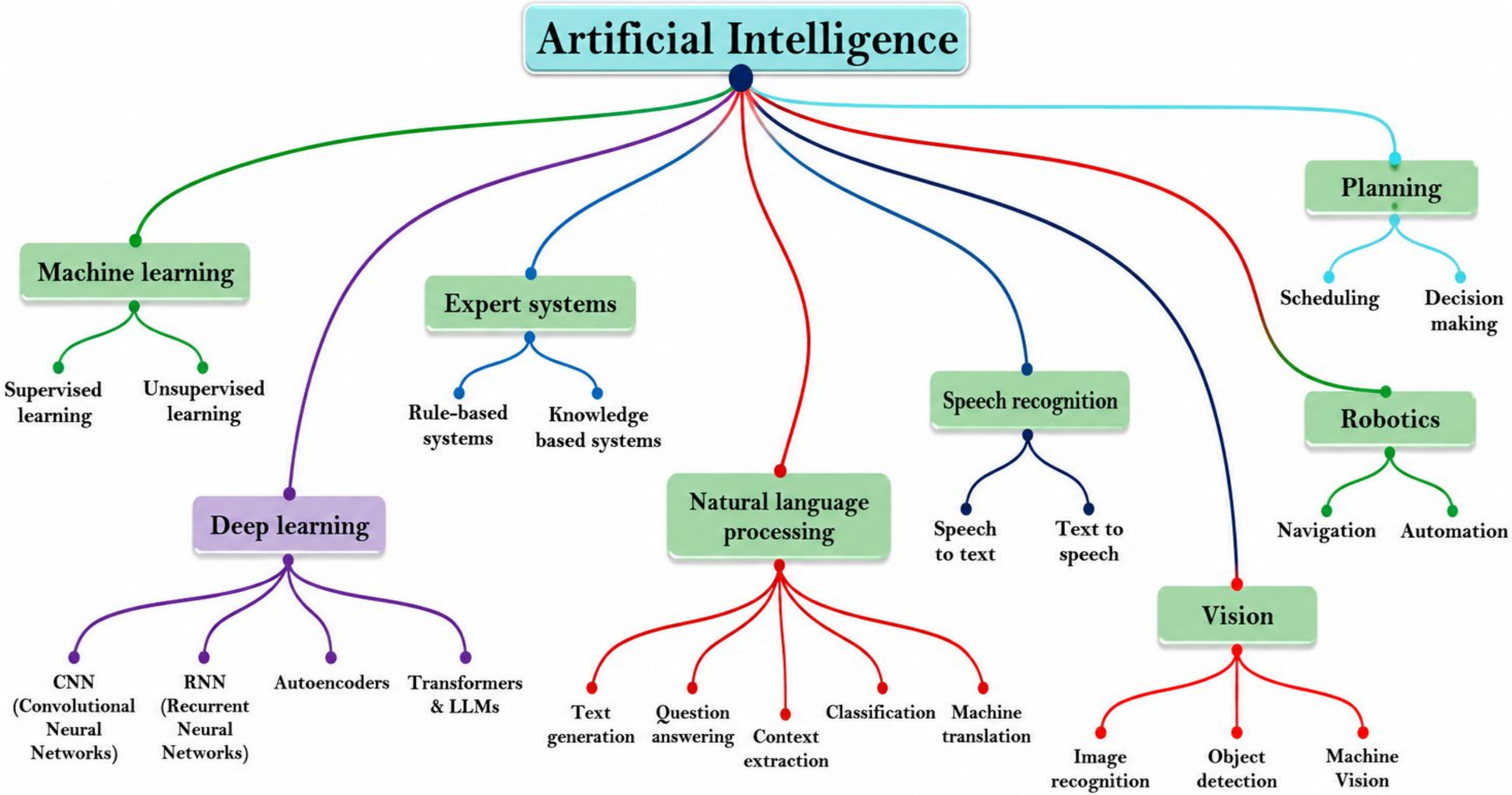
4. Speed : AI can given the information can be exact, **every time with speed**. But in HI is given the same information, **it cannot be as exact, and is slower.**

5. Memory : Humans are **unable to store large amount of data in the memory**. But in **Machine store large amounts of data and retrieve data fastely**.

Example : Chess Game.

Differences Between Human Intelligence And Artificial Intelligence

HUMAN INTELLIGENCE		ARTIFICIAL INTELLIGENCE
Natural intelligence (Born with)	Nature	Artificial / Machine-made (Created by humans)
Learn from experience, practice and observation	Learning	Learn from data, algorithms and training
Relatively slower in decision making	Speed	Very fast in processing and decision making
Gets tired and needs rest	Fatigue	Never gets tired (Can work 24/7)
May make mistakes due to emotions, distractions, etc.	Accuracy	More accurate and consistent (No emotions)
Limited memory (Forgets sometimes)	Memory	Huge memory (Stores large amount of data)
Highly creative and imaginative	Creativity	Low creativity (Works only on what it is given)
Examples: You, Me, Everyone	Examples	Examples: ChatGPT, Siri, Self-driving Cars, Robots, Alexa


Artificial Intelligence
Machine learning
Supervised learning
Unsupervised learning
Expert systems
Rule-based systems
Knowledge based systems
Deep learning
CNN (Convolutional Neural Networks)
RNN (Recurrent Neural Networks)
Autoencoders
Transformers & LLMs
Natural language processing
Text generation
Question answering
Context extraction
Classification
Machine translation
Speech recognition
Speech to text
Text to speech
Vision
Image recognition
Object detection
Machine Vision
Planning
Scheduling
Decision making
Robotics
Navigation
Automation

PREREQUISITES

To effectively learn **Deep Learning**, which is a **subset of machine learning** based on **Neural Networks** with **multiple layers**, it is beneficial to have a understanding of the following **fundamental concepts and topics**:

1. Mathematics:

- **Linear Algebra**: **Vectors, matrices, matrix operations, Eigen Values, Eigen Vectors, etc.**
- **Calculus**: **Derivatives, chain rule, gradients, partial derivatives, etc.**
- **Probability and Statistics**: **Probability distributions, Bayes' theorem, mean, variance, etc.**

PREREQUISITES

2. Programming Skills:

- **Python**: Deep learning frameworks such as **TensorFlow and PyTorch** are commonly used with Python.
- **NumPy**: For **numerical computing** in Python.
- **Pandas:** For **data manipulation and analysis**.
- **Jupyter Notebooks**: For **interactive coding** and **experimentation.**

3. Machine Learning Concepts:

- **Supervised Learning:** Understanding of concepts like **regression and classification.**
- **Unsupervised Learning: Clustering, Dimensionality reduction, etc.**
- **Model Evaluation: Cross-validation, overfitting, underfitting, etc.**

PREREQUISITES

4. Data Preprocessing:

- **Cleaning, normalization, feature scaling, handling missing data etc.**

5. Basic Understanding of Neural Networks:

- **Perceptron:** The basic building block of **neural networks.**
- **Activation Functions: Sigmoid, ReLU, tanh**, etc.
- **Back Propagation:** The **algorithm** used to **train neural networks.**

6. Deep Learning Frameworks:

- Familiarity with popular **deep learning libraries** such as **TensorFlow or Keras** is essential.

DEEP LEARNING

COURSE OUTCOMES

Upon successful completion of the course, the student will be able to:

CO1	**Understand** the fundamental **concepts, architectures, and applications of deep learning.**	Understand
CO2	**Apply** the **CNN and Deep Networks** to analyze **various architectures to solve Computer Vision Tasks.**	Apply
CO3	**Apply** Recurrent Neural Networks (RNNs) and Generative Adversarial Networks(GANs) to solve **natural language processing and image-to-image translation tasks**	Apply
CO4	**Analyze** the performance of deep learning architectures, including CNNs, RNNs and GANs for various tasks in **computer vision, natural language processing and generative modeling.**	Analyze

DEEP LEARNING

	Test	Test Number	Marks
CIE	**Objective Exam (10)**	Objective Exam -1	**10**
		Objective Exam -2	**10**
	Assignment (5)	Assignment -1	**5**
		Assignment – 2	**5**
	Descriptive Exam (15)	Descriptive Exam – 1	**15**
		Descriptive Exam - 2	**15**

	CO	Marks
SEE	CO1	14
	CO2	14
	CO3	21
	CO4	21



Syllabus		
Unit No	**CONTENTS**	**Mapped CO**
1	**Fundamentals of Deep Networks** – Biological and Machine Vision, Human and Machine Language, Introduction to Deep Learning, Anatomy of Neural Network, Artificial Neural Networks. **Common Architectural Principles of Deep Networks**: Parameters, Layers, Activation Functions, Loss Functions, Hyperparameters, how deep learning works, Types of Deep Learning, Applications of Deep Learning.	**CO1**
2	**Building Blocks of Deep Networks** – Boltzmann Machine, Restricted Boltzmann Machine, Autoencoders, Variational Autoencoders. **Major Architectures of Deep Networks:** Deep Belief Networks, Generative Adversarial Networks.	**CO1, CO2, CO4**
3	**Convolutional Networks (CNNs)** – Motivation, the Convolution Operation, Convolutional Layers, Pooling, Structured Outputs, LeNet, AlexNet, VGG Net.	**CO1, CO2, CO4**
4	**Recurrent Neural Networks:** Why RNN, Architecture of RNN, Types of RNN, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, and the Long Short-Term Memory - Gated Recurrent Units, Bidirectional LSTMs, and Bidirectional RNNs. **Optimization Algorithms:** Optimization and Deep Learning, Adagrad,adam.	**CO1, CO3,CO4**
5	**Recent Trends of Deep Learning**: What are Transformer Models, Architecture of Transformers. **Applications:** Computer Vision, Speech Recognition, Natural Language Processing.	**CO1, CO3, CO4**

Learning Resources

Text Books

1. **Deep Learning- Ian Goodfellow, Yoshua Bengio and Aaron Courvile, MIT Press, 2016**
2. **Deep Learning with Python - Francois Chollet, Released December 2017,** Publisher(s): Manning Publications, ISBN: 9781617294433
3. **Deep Learning Illustrated:** A Visual, Interactive Guide to Artificial Intelligence - **Jon Krohn, Grant Beyleveld, Aglaé Bassens, Released September 2019,** Publisher(s): Addison-Wesley Professional, ISBN: 9780135116821
4. **Deep Learning from Scratch - Seth Weidman, Released September 2019,** Publisher(s): O'Reilly Media, Inc., ISBN: 9781492041412

Reference Books

1. Artificial Neural Networks, Yegnanarayana, B., PHI Learning Pvt. Ltd, 2009.

2. Matrix Computations, Golub, G.,H., and Van Loan,C.,F, JHU Press, 2013.

3. Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004.

E-Resources & other digital material
1. Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc22_cs22/preview'
2. https://www.datacamp.com/tutorial/how-transformers-work
3. https://d2l.ai/index.html

Introduction to Deep Learning

- **Deep learning** is a **subset of machine learning** in which a model **learns to perform classification tasks directly from images, text, or sound.** Deep learning is usually implemented using a **Neural Network Architecture.**
- **Father of Deep Learning**, Geoffrey Hinton.
- Deep learning is a **type of artificial intelligence** that can **"duplicate" the human brain's functioning.**
- The term **"deep"** refers to the **more number of hidden layers** in the network.
- **Traditional neural networks** contain **only 2 or 3 layers**, while **deep networks can have hundreds**.
- It uses **Artificial Neural Networks (ANNs)** with **multiple hidden layers** to **automatically learn features and patterns from large amounts of data.**

Introduction to Deep Learning

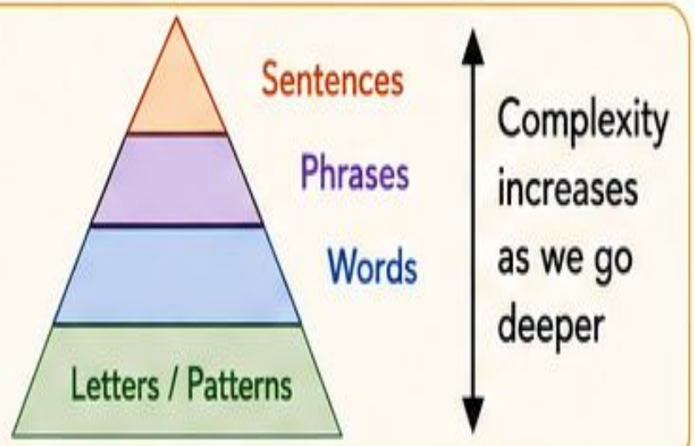
- **Deep learning models** can **recognize complex patterns** in **pictures, text, sounds, and other data** to produce accurate **insights and prediction**. It improves the ability to **classify, recognize, detect and describe using data.**
- **Neural networks** are **made up of layers** of **interconnected nodes,** and **each node is responsible for learning a specific feature of the data.**
- **For Example**, we have to find out the **sentences**- the **first layer of nodes** might learn to **identify the letters**, **the second layer** might learn to **identify the words**, and **the third layer** might learn to **identify sentences** etc. Like this we **may increase the hidden layers** in the Artificial Neural Network.

Deep Learning



In Simple Words:

Deep Learning models act like a learning pyramid. The lower layers learn simple features, and as we go deeper, the layers learn more complex and meaningful features, until the final layer can understand the complete input (like a sentence).

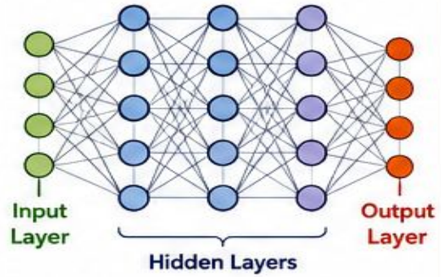


Key Points of Deep Learning

Feature	Description
Deep Architecture	Uses multiple hidden layers
Automatic Feature Learning	Learns features automatically from raw data
Hierarchical Learning	Learns from simple features to complex patterns
Large Data Requirement	Performs best with large datasets
High Accuracy	Excellent performance on complex tasks
End-to-End Learning	Learns directly from input to output
Self-Learning	Improves through backpropagation and optimization
Handles Unstructured Data	Works with images, text, speech, and videos
High Computational Power	Requires GPUs/TPUs for efficient training
Good Generalization	Makes accurate predictions on unseen data

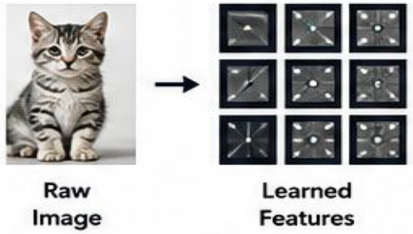
KEY FEATURES OF DEEP LEARNING

1 MULTIPLE HIDDEN LAYERS



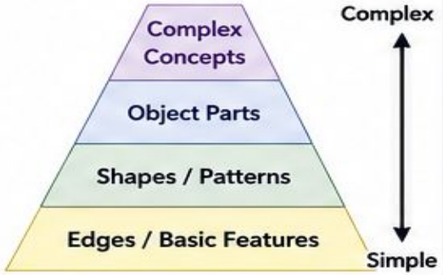
Uses many hidden layers (deep architecture) between input and output layers.

2 AUTOMATIC FEATURE EXTRACTION



Automatically learns important features from raw data without manual feature engineering.

3 HIERARCHICAL LEARNING



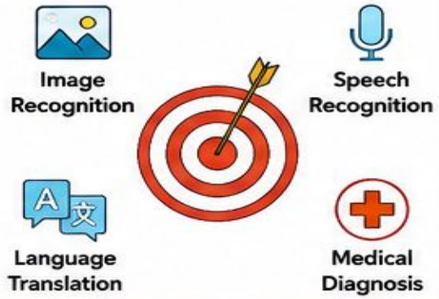
Learns from simple features in lower layers to complex patterns in deeper layers.

4 LEARNS FROM LARGE DATASETS



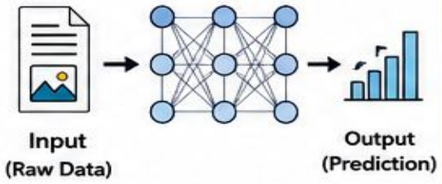
Performs better when trained on large amounts of data. More data → Better accuracy.

5 HIGH ACCURACY



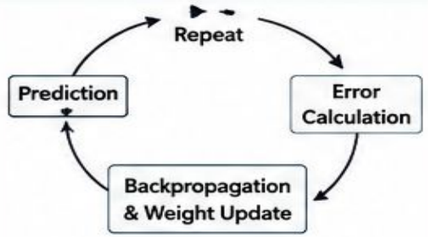
Provides excellent accuracy in complex tasks like vision, speech, NLP, and more.

6 END-TO-END LEARNING



Learns directly from input to output without manually designing intermediate steps.

7 SELF-IMPROVING THROUGH TRAINING



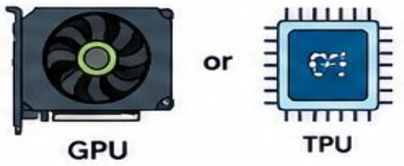
Continuously improves by adjusting weights using backpropagation and optimization algorithms.

8 HANDLES UNSTRUCTURED DATA



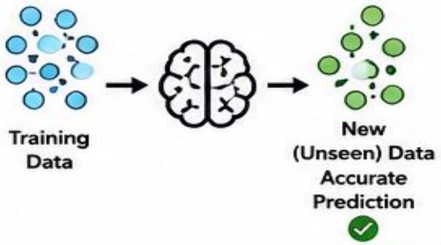
Works effectively with unstructured data such as images, audio, text, and videos.

9 REQUIRES HIGH COMPUTATIONAL POWER



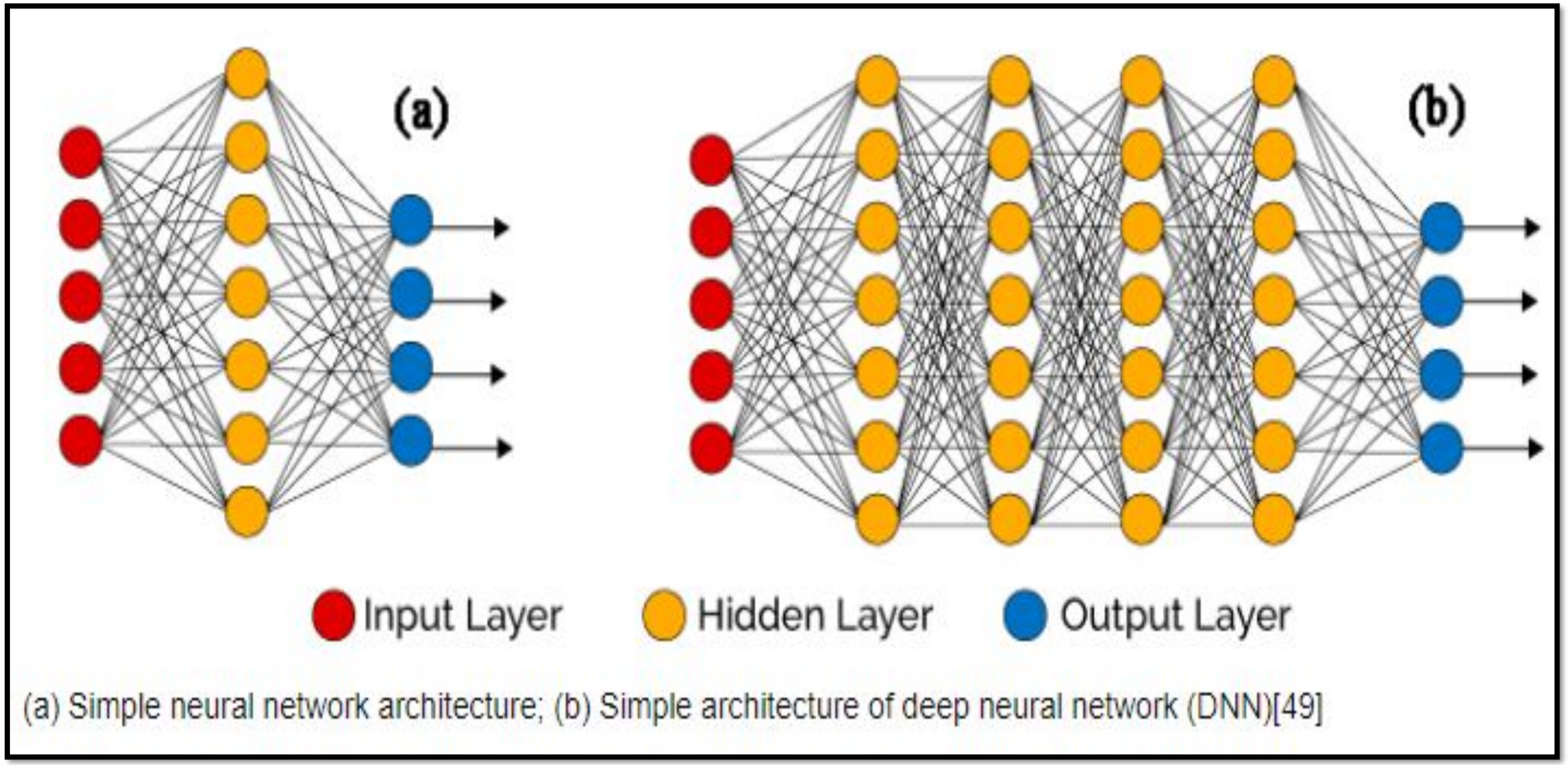
Needs powerful hardware like GPUs or TPUs. Training may take hours or even days.

10 GOOD GENERALIZATION ABILITY



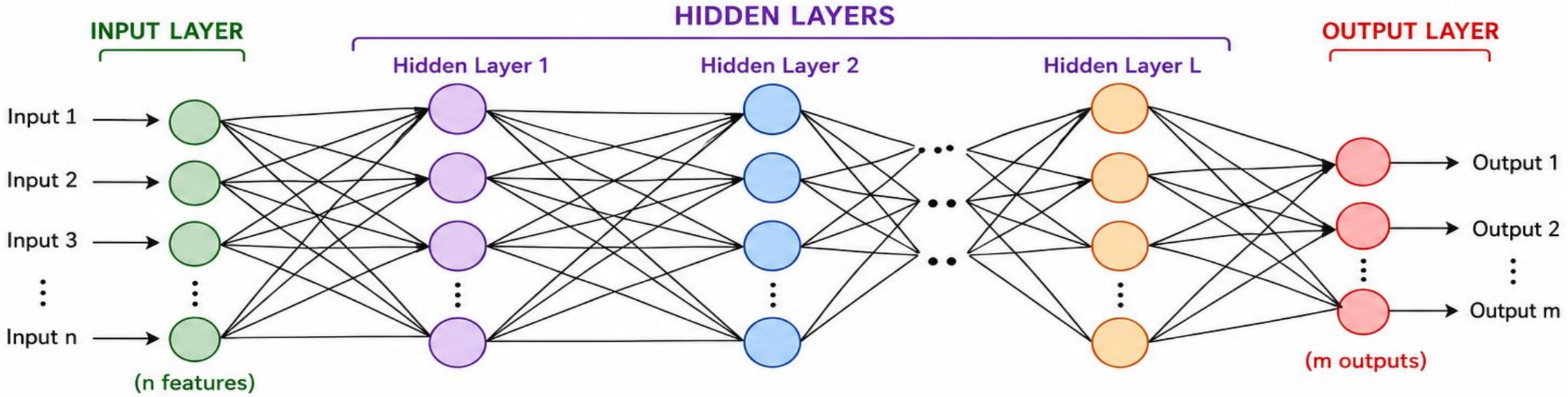
Can accurately predict outcomes for new, unseen data, not just the training data.

SUMMARY: Deep Learning uses deep neural networks to automatically learn hierarchical features from large amounts of data, achieve high accuracy, and solve complex real-world problems.



(a) Simple neural network architecture; (b) Simple architecture of deep neural network (DNN)[49]

Architecture of Deep Neural Network (DNN)

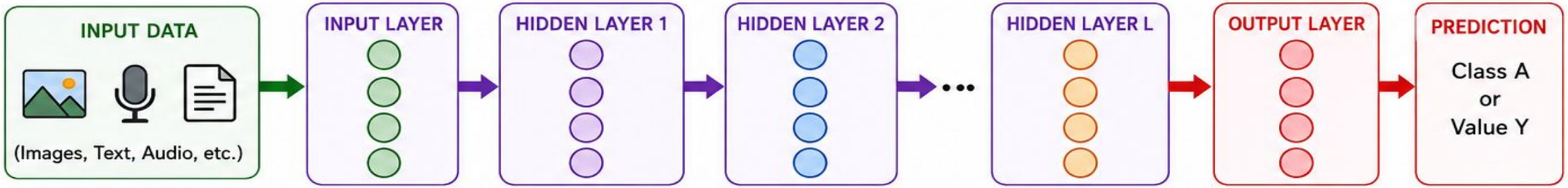


INPUT LAYER
Receives the input data and forwards it to the next layer.

HIDDEN LAYERS
Each hidden layer performs computations and learns features. Early layers learn simple features; deeper layers learn complex features.

ACTIVATION FUNCTION
Each neuron applies an activation function (ReLU, Sigmoid, tanh, etc.) to introduce non-linearity.

OUTPUT LAYER
Produces the final prediction or classification.



More hidden layers → Deeper the network → Better feature learning → Higher accuracy

➢ Differences between AI, ML and DL

Artificial Intelligence (AI)



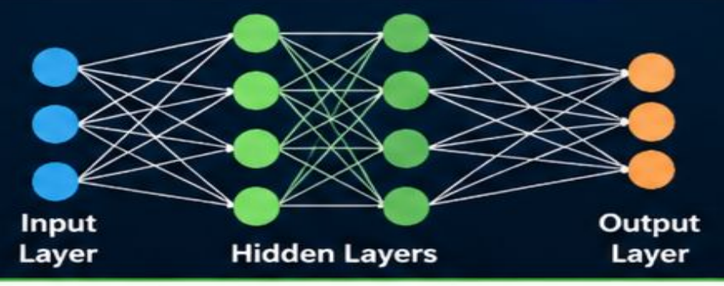
- AI is the **broadest field.**
- **Goal:** Create intelligent machines.
- Includes ML, DL, Robotics, Expert Systems, etc.
- **Not always learns from data.**
- Uses **logic, rules, reasoning.**
- **Example:** Chess program, Siri, Self-driving car.

Machine Learning (ML)



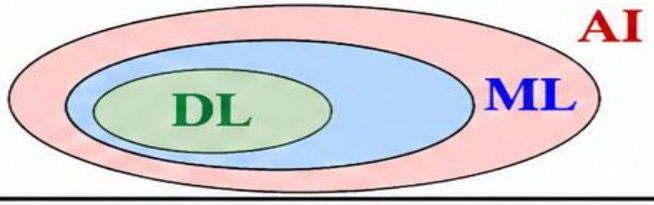
- ML is a **subset** of AI.
- **Goal:** Learn from data and improve performance.
- Uses statistical methods and algorithms.
- **Learns from data (experience).**
- Needs **feature engineering.**
- **Example:** Email spam detection, Recommendations.

Deep Learning (DL)



- DL is a **subset** of ML.
- **Goal:** Learn complex patterns using neural networks.
- Uses deep neural networks (many layers).
- **Learns automatically from large data.**
- No need for manual **feature engineering.**
- **Example:** Face recognition, ChatGPT, Image classification.

➢ **AI** is the **Science**, **ML** is the **Approach** using **data**, and **DL** is the **Powerful Technique** using **Neural Networks.**



From Human Learning to Deep Learning

The Journey of Intelligence

Human

I can learn everything automatically from experiences. Can u learn?

Machine

Yes, I can also learn from past data with the help of Machine learning



I can learn from experience and examples, but it's hard for me to learn very complex patterns in large amounts of data. Can u learn better?

Deep Learning

Yes! I use **Deep Learning** (a part of Machine Learning) which uses **deep neural networks** to automatically learn very complex patterns from huge amounts of data like images, text, audio and video.



Relationship

Artificial Intelligence (AI)
The science of making machines intelligent

→

Machine Learning (ML)
A subset of AI that learns from data and improves over time

→

Deep Learning (DL)
A subset of ML that uses deep neural networks to learn complex patterns from large data

Applications

- Email Spam Detection
- Face Recognition
- Speech Recognition
- Image / Video Analysis
- Autonomous Driving
- Medical Diagnosis

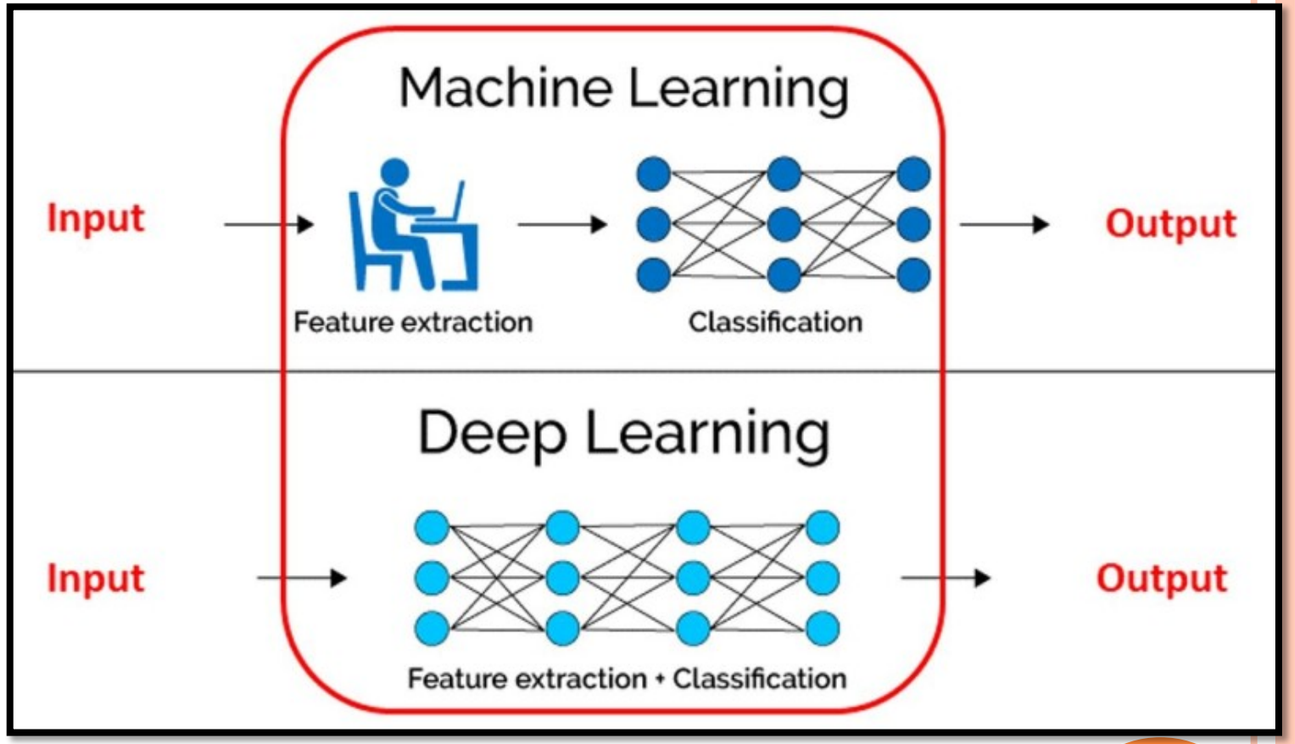
Conclusion

Deep Learning is a powerful part of **Machine Learning**, which is part of **Artificial Intelligence**.
Together, they help machines learn, reason and solve real-world problems.

Imagine teaching a child to identify a dog.

- **Machine Learning:** You tell the child the features:
  - Four legs
  - Tail
  - Fur
  - Ears
- **Deep Learning:** You simply show the child **thousands of dog images**. The system automatically learns the important features without being explicitly told.

Deep learning **eliminates some of data pre-processing** that is typically **involved with machine learning**.



MACHINE LEARNING

Features are extracted using algorithms, then given to ML model for learning.

INPUT

Image of a dog



FEATURE EXTRACTION

Algorithms extract features from the image

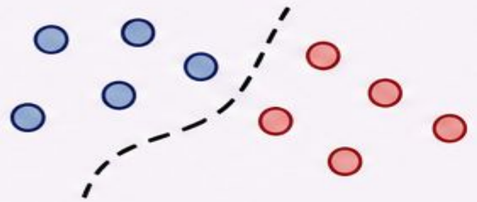
Feature Extraction Algorithms

Extracted Features

- Ears shape
- Eye position
- Nose texture
- Fur color
- Body shape
- Legs count
- etc.

LEARNING (CLASSIFICATION)

ML Algorithm learns using extracted features



ML Algorithm

OUTPUT

Dog

(Prediction)

In Machine Learning, features are extracted using algorithms, then given to ML model.

DEEP LEARNING

The model (neural network) automatically learns features and performs classification.

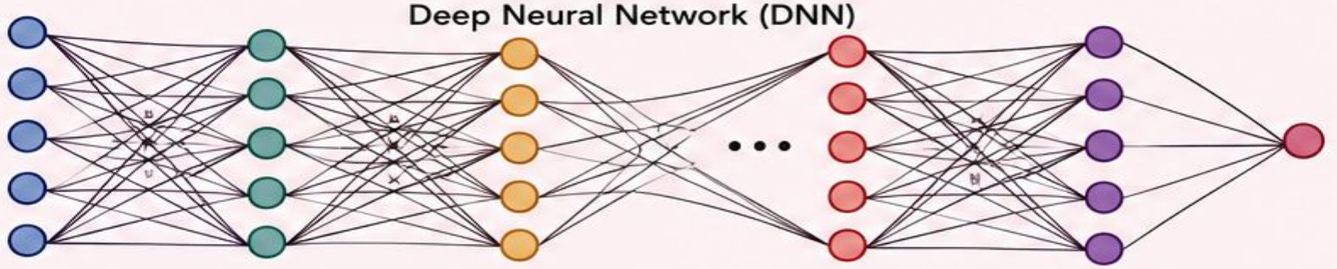
INPUT

Image of a dog



FEATURE LEARNING + CLASSIFICATION

Deep Neural Network (DNN)



Learns edges and textures

Learns shapes (parts of dog)

Learns complex patterns

Recognizes dog as a whole

OUTPUT

Dog

(Prediction)

In Deep Learning, the model automatically learns features from raw data and does the classification.

Feature	Machine Learning (ML)	Deep Learning (DL)
1. Introduction	Developed in the 1950s–1980s	Popular since the 2010s with big data & GPUs
2. Learning Method	Learns patterns using ML algorithms	Learns patterns using deep neural networks
3. Feature Extraction	Features are extracted using algorithms (often with human guidance/domain knowledge)	Features are learned automatically by the neural network
4. Data Requirement	Works well with small to medium datasets	Requires very large datasets
5. Best For	Structured/tabular data	Images, speech, text, video, and other unstructured data
6. Model Complexity	Simple or moderately complex models	Very deep models with many hidden layers
7. Training Time	Faster training	Longer training
8. Hardware	CPU is usually sufficient	GPU/TPU is preferred
9. Accuracy	Good for simple problems	Better for complex problems
10. Interpretability	Easier to understand and explain	Harder to explain (Black-box model)
11. Examples	Spam detection, House price	Face recognition, ChatGPT, Self-driving cars, Medical image diagnosis, Speech

Reasons to use Deep Learning

1. **Analyzing Unstructured Data:** Deep learning algorithms can be **trained to look at text data** by **analyzing social media posts, news, and surveys** to provide **valuable business** and customer insights.
2. **Feature engineering:** A deep learning algorithm can **save time** because it does **not require humans to extract features manually** from **raw data.**
3. **Efficiency:** When a deep learning Algorithm is properly trained, it can perform **thousands of tasks over and over again**, **faster than humans**.

4. Training: The **Neural Networks** used in **deep learning** have the ability to be applied to **many different data types and applications.** Additionally, a deep learning model can **adapt by retraining it with new data.**

DL is usually a **more complex and high-performance GPU** to **analyze all images.**

5. Handling large and complex data: Deep Learning algorithms can deal **with large and complex datasets** that would be challenging for **traditional Machine Learning algorithms** to handle. This makes **it useful for finding** insights from **big data**, such as **posts on social media, webpages, videos, audio files,** and **sensor data.**

Reasons to use Deep Learning

6. Handling Non-Linear Relationships: This allows it to model **complex phenomena and capture higher-level abstractions**. Some of the examples of handling non-linear relationships by Deep Learning algorithms are:

a) **Physical Systems:** A **Deep Potential Molecular Dynamics (DPMD)** achieved an **accuracy of 98.9%** on **predicting** the **potential energy surface of water molecules**, which is a **highly non-linear function of atomic positions.**

b) **Biological Systems:** A **Deep Learning model** called **DeepChrome** achieved an **accuracy of 89.9%** on predicting the **chromatin state of genes.**

How Deep Learning Works

- Deep learning work on **Neural Network Architectures**
- Deep learning is implemented by the help of **deep networks,** which are nothing but **neural networks** with **multiple hidden layers.**
- It **came in the 1980s**, but due to a **lack of labeled data and computing power, i**t was not **popular at that time.**
- Today, we have **large amounts of labeled data**. For example, a **driverless car** requires **millions of images and thousands of hours of video to train** with **high accuracy**.
- The **number of hidden layers** in the **neural network** usually refers to **"deep".** Hidden layers in **deep neural networks** can have **as many as 150.**

How Deep Learning Works

- Deep learning work on **Neural Network Architectures**
- Deep learning is implemented by the help of **deep networks,** which are nothing but **neural networks** with **multiple hidden layers.**
- It **came in the 1980s**, but due to a **lack of labeled data and computing power, i**t was not **popular at that time.**
- Today, we have **large amounts of labeled data**. For example, a **driverless car** requires **millions of images and thousands of hours of video to train** with **high accuracy**.
- The **number of hidden layers** in the **neural network** usually refers to **"deep".** Hidden layers in **deep neural networks** can have **as many as 150.**

HOW DEEP LEARNING WORKS

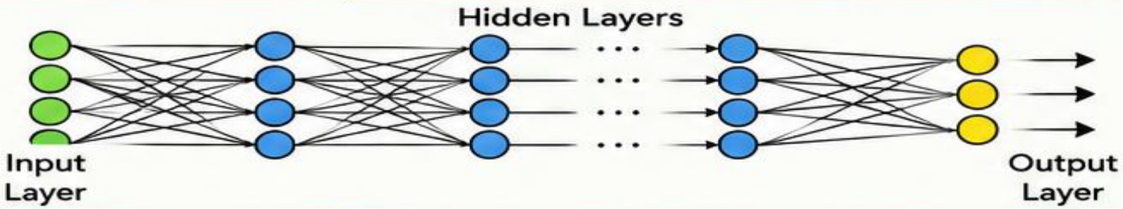
1 LARGE DATA

Huge amount of labeled data



2 DEEP NEURAL NETWORK

Many hidden layers learn from data



3 AUTOMATIC FEATURE LEARNING

The model automatically learns useful features



4 PREDICTION

Model uses learned features to make predictions



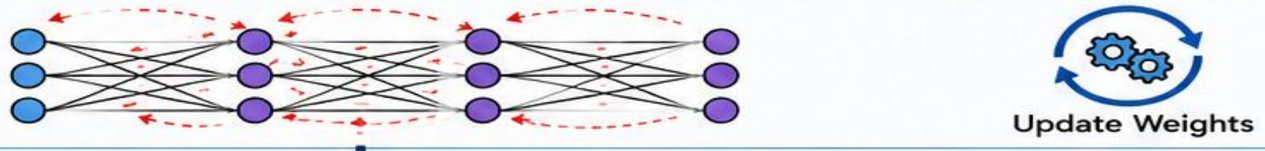
5 ERROR CALCULATION

Compare prediction with actual label



6 WEIGHT UPDATE

Use backpropagation to update weights and reduce error



7 BETTER PREDICTION

Model gets better and makes more accurate predictions



1. Here, we are passing the high dimensional data to the input layer. To match the dimensionality of the input data, the input layer will contain multiple sub-layers of perceptrons so that it can consume the entire input.

2. The output received from the input layer will contain patterns and will only be able to identify the edges of the images based on the contrast levels.

3. This output will be fed to the Hidden layer 1 where it will be able to identify various face features like eyes, nose, ears etc.

4. Now, this will be fed to the hidden layer 2 where it will able to form the entire faces. Then, the output of layer 2 is sent to the output layer.

5. Finally, the output layer performs classification based on the result obtained from the previous and predicts the name.

HOW DEEP LEARNING WORKS

INPUT

Raw Data



(Example: Chest X-ray Image)

Hidden Layer 1 (Simple Features)

Learns edges, lines, and basic patterns

Hidden Layer 2 (Intermediate Features)

Learns shapes, textures, and combinations

Hidden Layer 3 (Complex Features)

Learns high-level patterns and structures

Hidden Layer 4 (More Abstract Features)

Learns more abstract representation of the object

OUTPUT

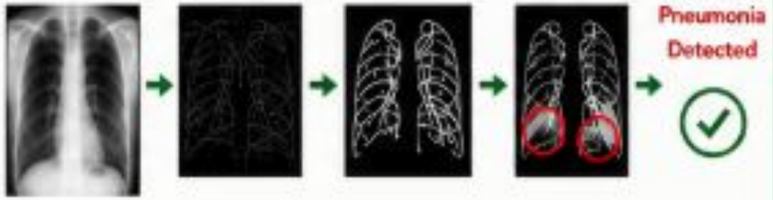
Prediction

Pneumonia Detected

EXAMPLES OF DEEP LEARNING IN REAL LIFE

1. MEDICAL IMAGE DIAGNOSIS

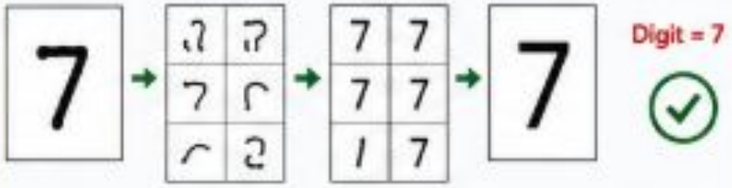
Input: Chest X-ray → Layer 1: Edges → Layer 2: Lung Patterns → Layer 3: Abnormal Regions → Output: Pneumonia Detected



The model learns step-by-step from simple edges to complex patterns and detects pneumonia.

2. HANDWRITTEN DIGIT RECOGNITION

Input: Image of Digit → Layer 1: Edges & Curves → Layer 2: Parts of Digit → Layer 3: Digit Recognized → Output: Digit = 7



The model learns edges → parts → complete digit and recognizes the number.

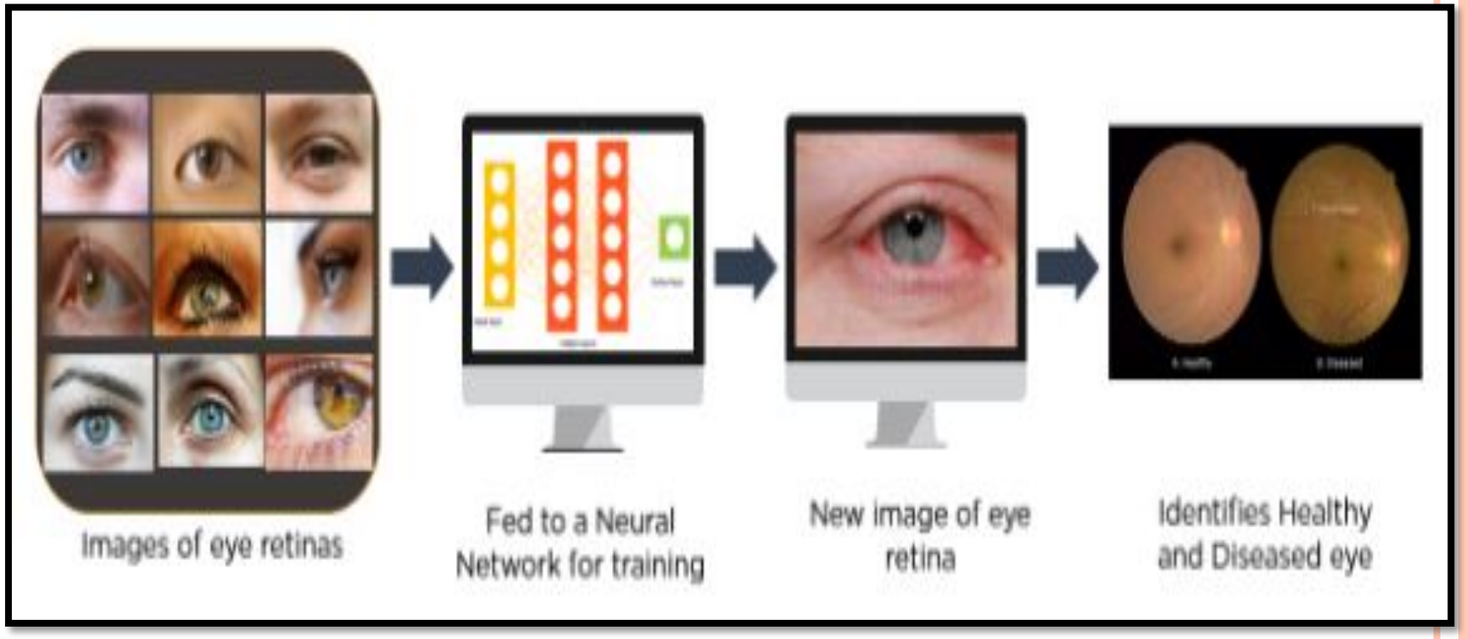
3. FACE RECOGNITION

Input: Face Image → Layer 1: Edges & Contours → Layer 2: Facial Features → Layer 3: Feature Combination → Output: Person Recognized

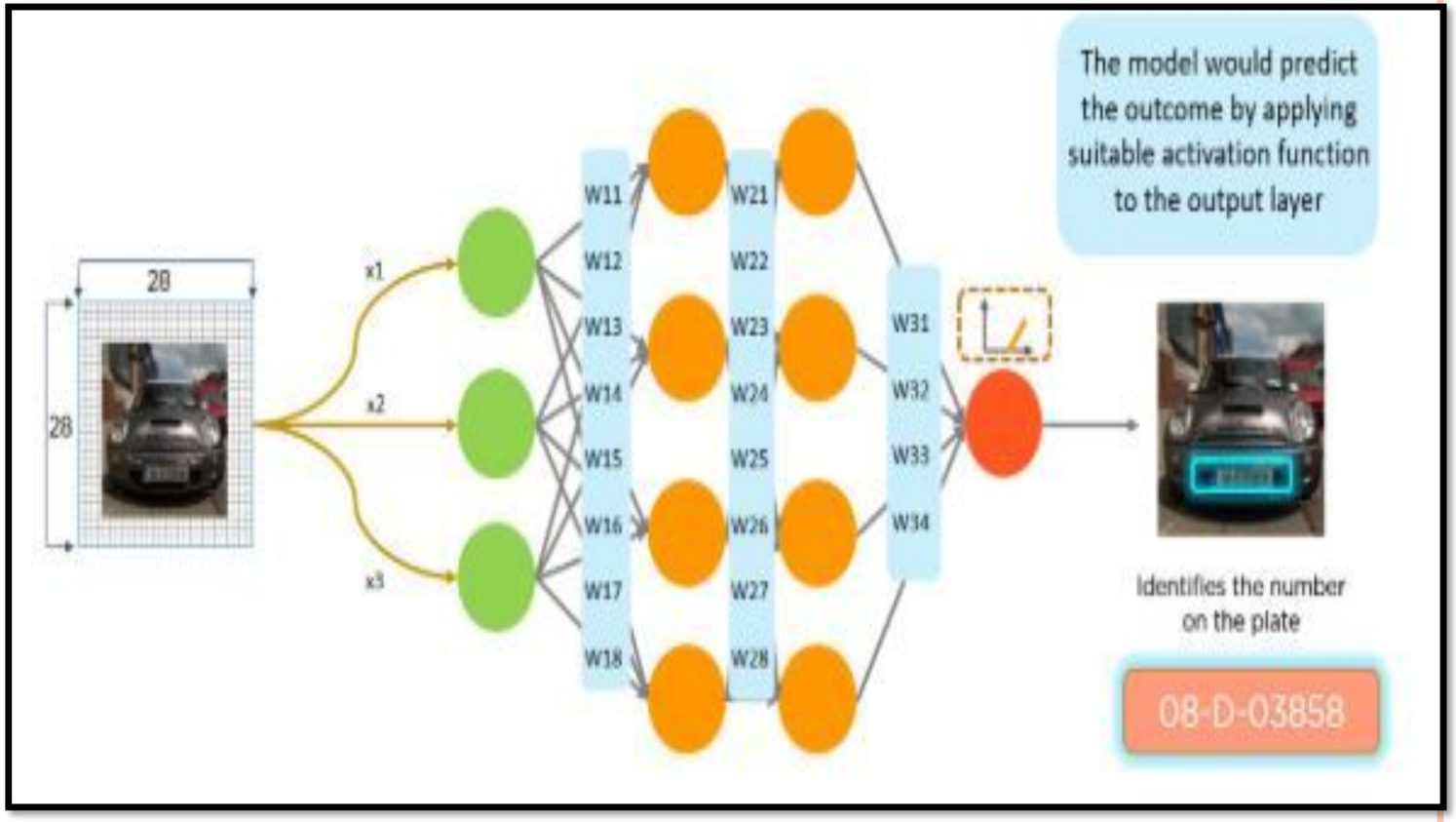


The model detects edges → features → combines them and recognizes the person.

Here is an example of a **Deep Learning** and **neural network** that **uses large sets of unlabeled data of eye retinas.** The network model is trained on this data to find out whether or not a person has **diabetic retinopathy.**



Here is an example of a **Deep Learning** and **neural network** are used to **identify the number on a license plate.** This technique is used by **many countries to identify rules violators and speeding vehicles.**



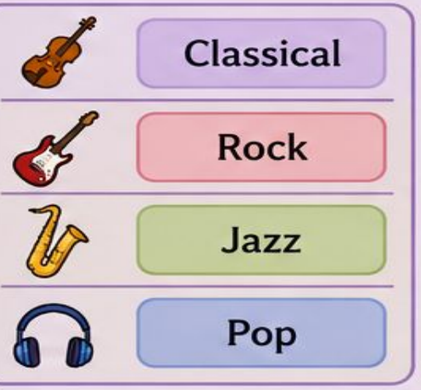
How Deep Learning Works (At a High Level)

Deep learning to non-experts using a music genre classification example

Lots of labeled data

Many music recordings
+
"This is Classical",
"This is Rock",
"This is Jazz",
"This is Pop"





Deep neural network

Many stacked layers automatically learn
(from sounds → patterns → meaning)
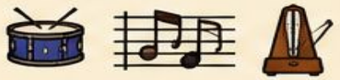
Layer 4: Full song representation (genre understanding)

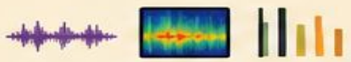


Layer 3: High-level representation (melody, harmony, instruments)



Layer 2: Music elements (rhythm, pitch, tempo)





Layer 1: Basic patterns (waveforms, frequencies)

Prediction / Decision

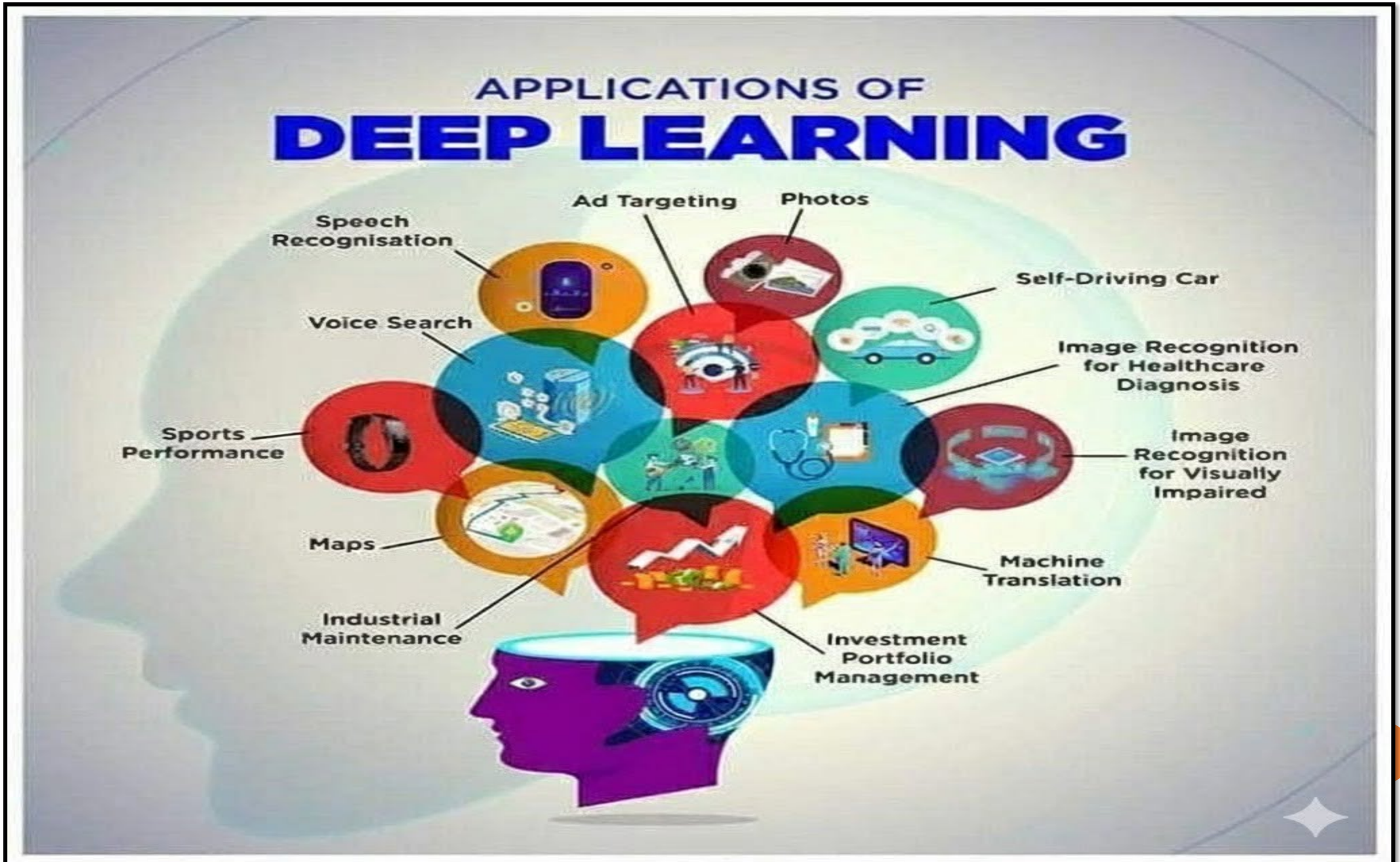
What genre is this song?

Rock	94%
Classical	3%
Jazz	2%
Pop	1%

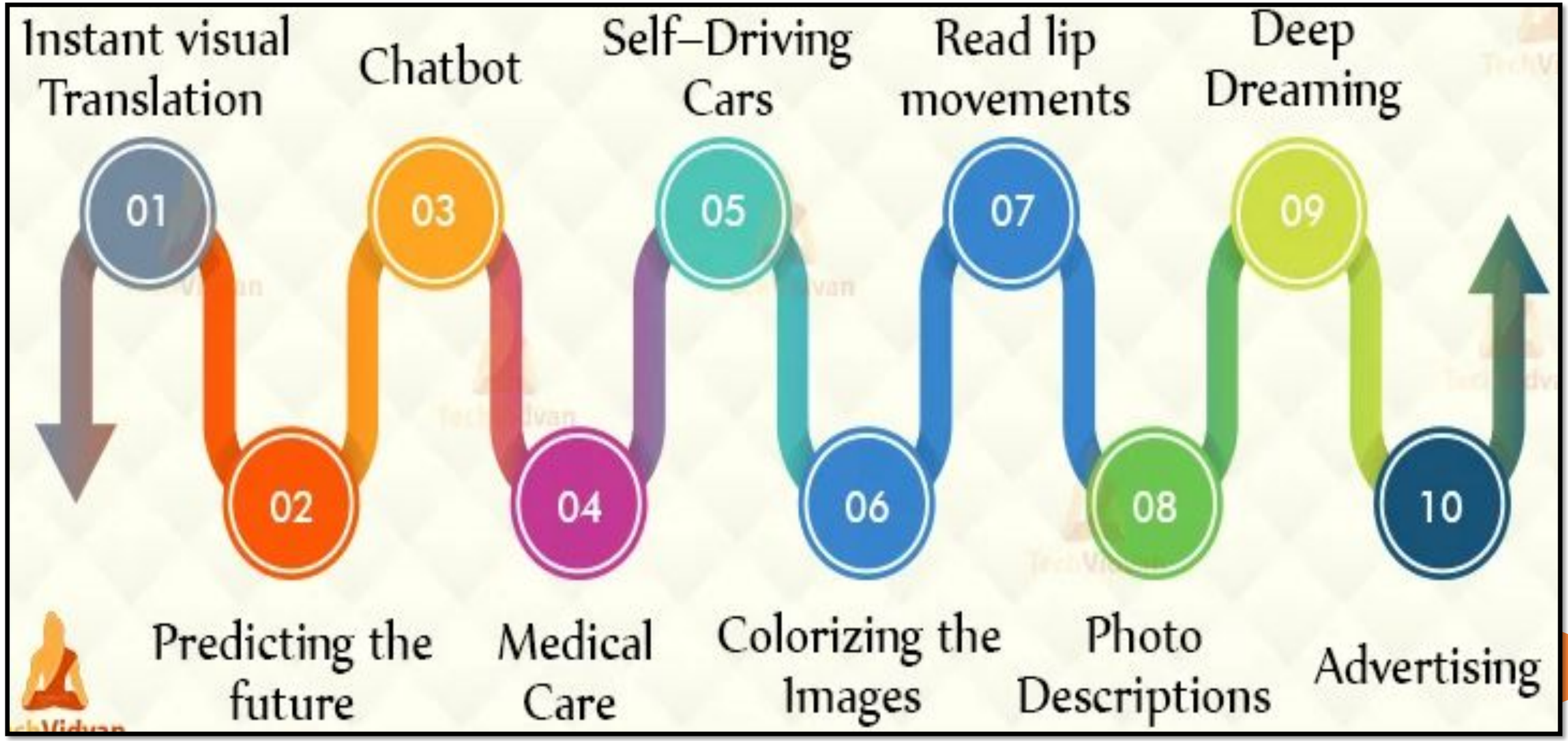
Final decision: **Rock** ✓



Deep = many layers, not deep thinking


APPLICATIONS OF
DEEP LEARNING
Ad Targeting
Photos
Speech Recognisation
Self-Driving Car
Voice Search
Image Recognition for Healthcare Diagnosis
Sports Performance
Image Recognition for Visually Impaired
Maps
Machine Translation
Industrial Maintenance
Investment Portfolio Management

Deep Learning Applications



CONTD..

1. Instant Visual Translation: With deep learning, **identification of text on the images is possible**. Once identification completes, it **translates the text immediately and recreates the image with translated text**.

For example, Suppose you **visit an unknown country** whose **local language is not known to you**. An app like **google translator converts the text of an image in an understandable** language.

With the power of deep learning, **Neural Machine Translation (NMT)** has arisen as the **most powerful algorithm** to perform this task.

This conversion is composed by using Two Recurrent Neural Networks(RNN)

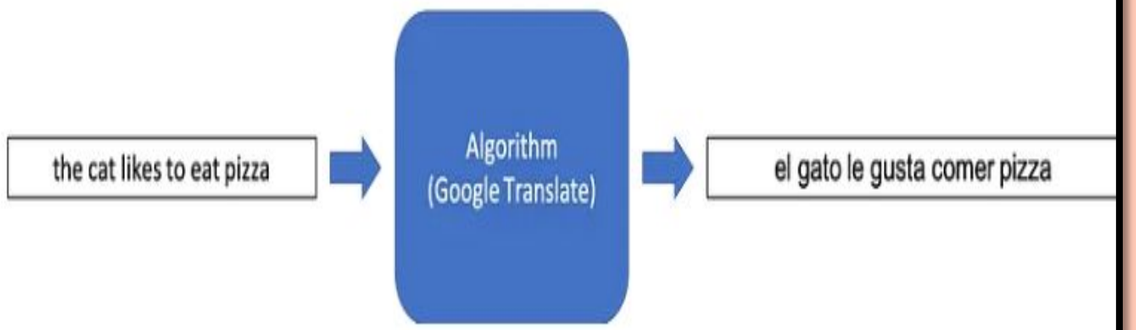
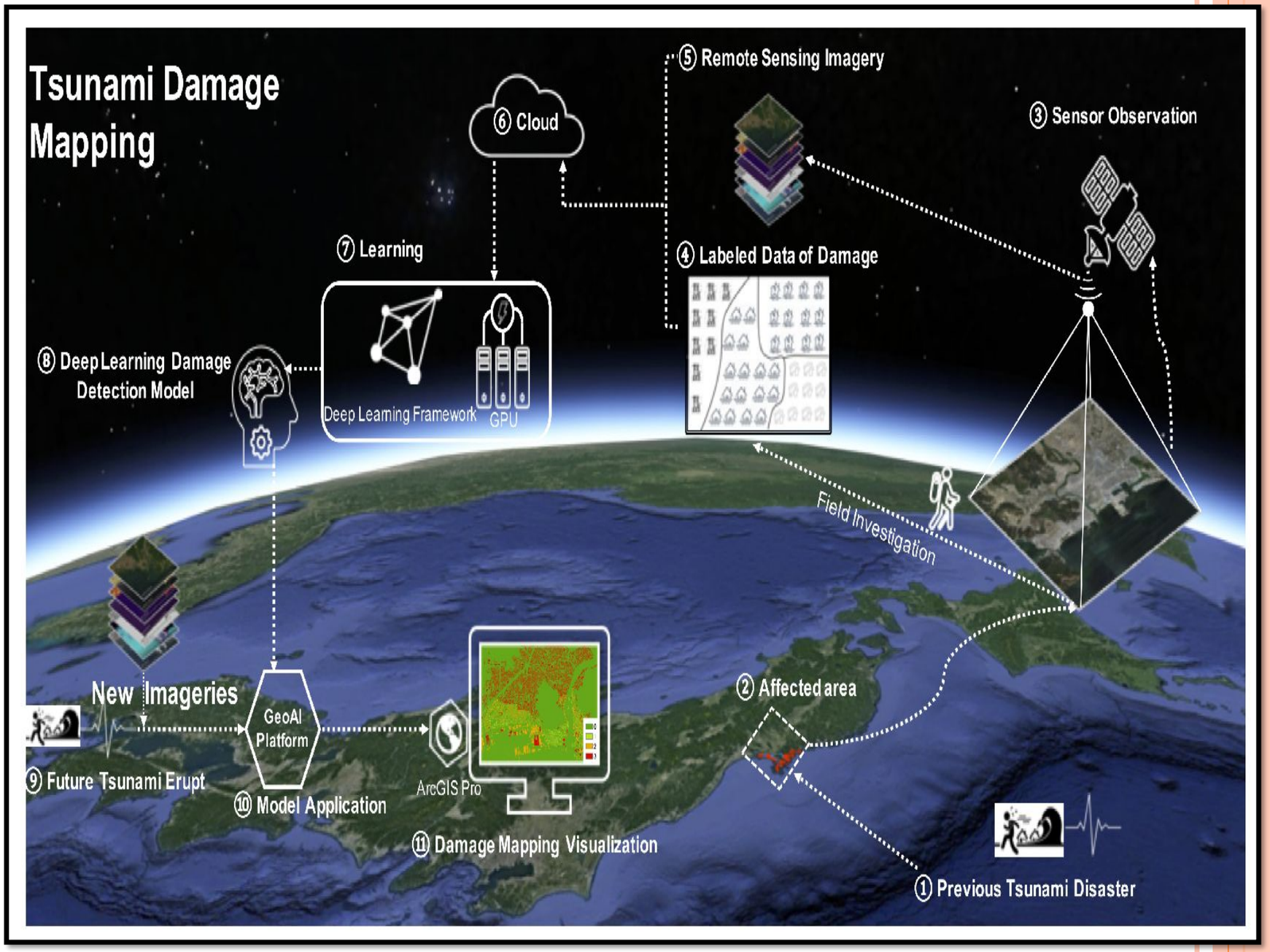


Figure 1: Translation from English to Spanish of the English sentence "the cat likes to eat pizza"

CONTD..

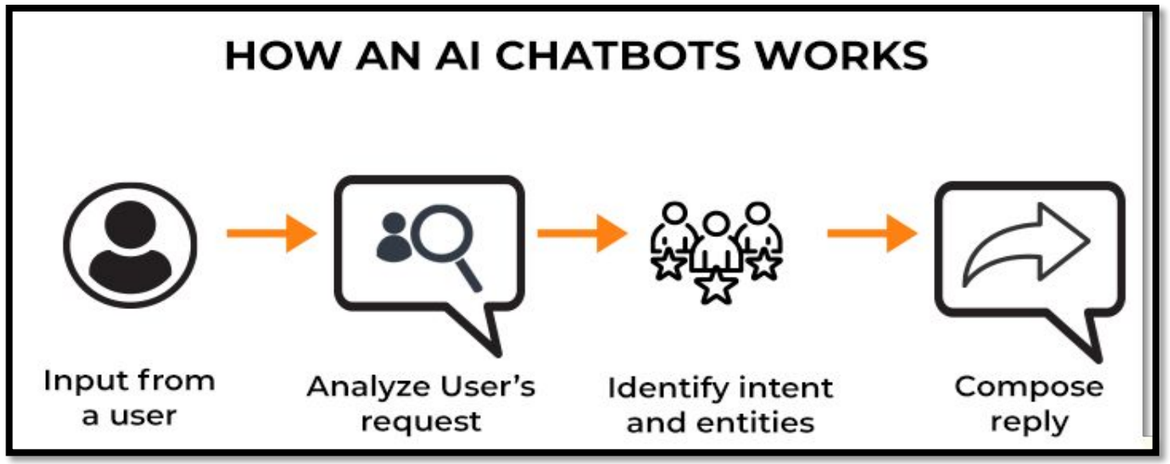
2. Predicting the Future: Deep neural networks help in the **prediction of earthquakes, tsunamis, cyclones, etc.** So that preventive measures can be taken to **save lives from falling into the grasp of natural calamities. This can be done by using Deep Neural Networks and Conversions can be done by Auto Encoders.**

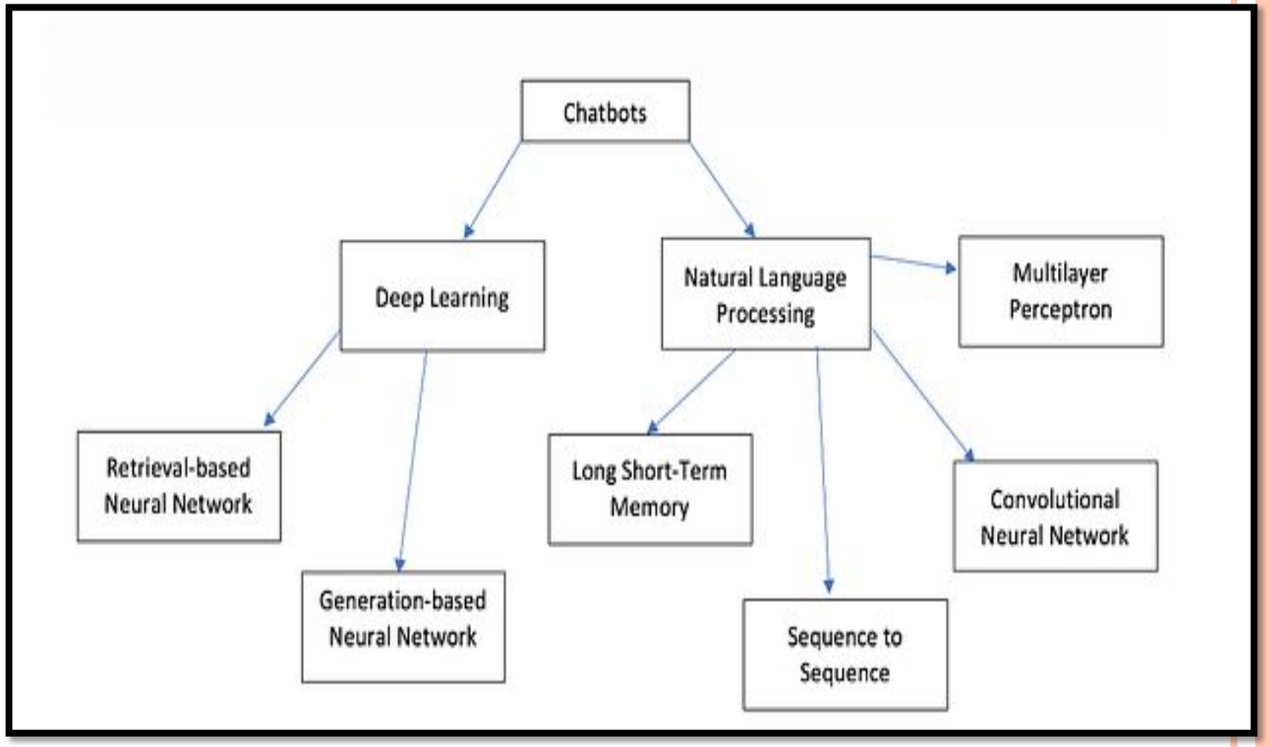


3. Chatbot: With the help of **deep learning Applications**, **chatbots** are becoming smarter day by day.

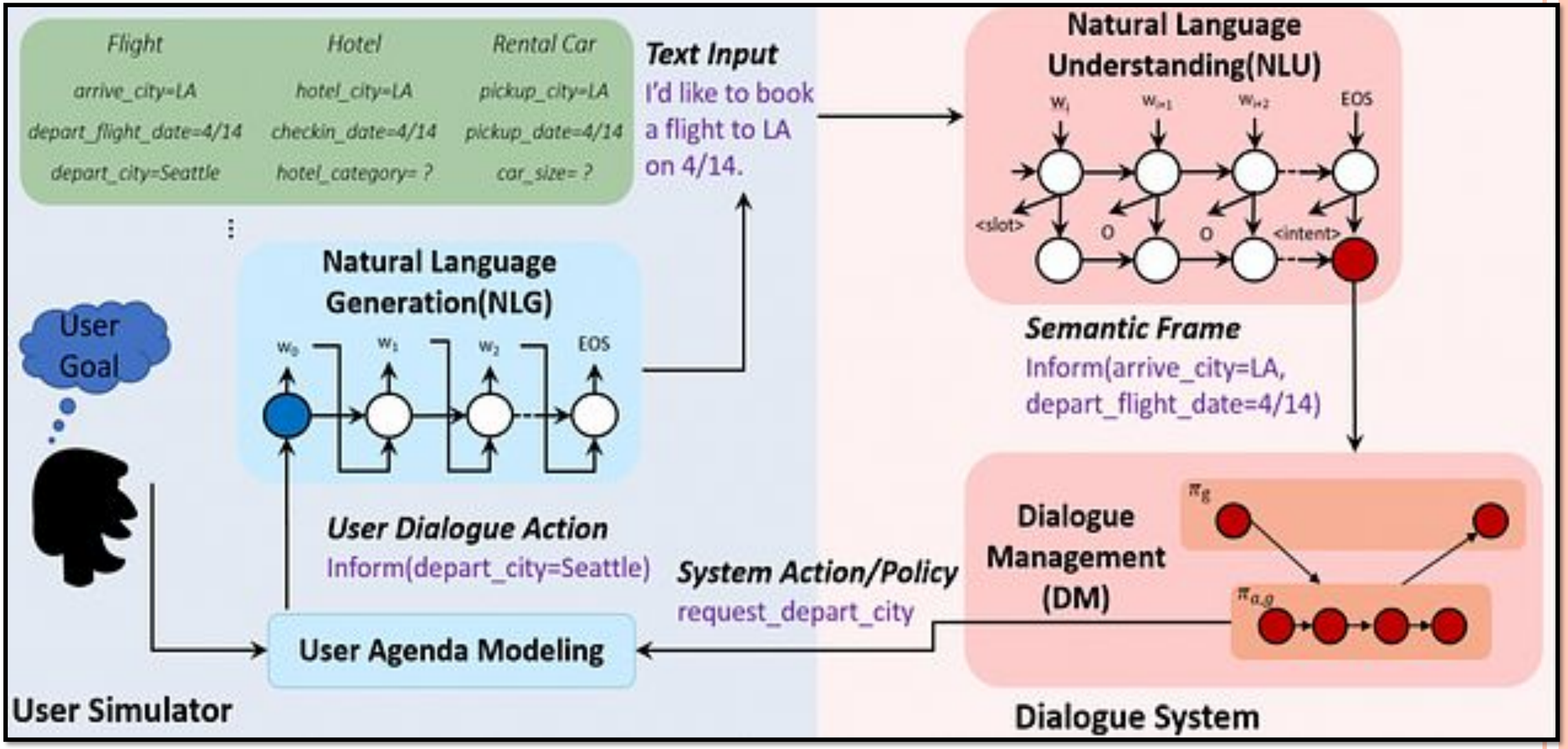
- **Amazon, Flipkart and many e-commerce websites** are **using chatbots for customer services**.
- **Transportation apps** like **Ola and Uber** are also **implementing chatbots to provide personalized aid**.
- **Siri** is also a good example of a **chatbot.**
- **Chatbots** work by **adopting mainly 3 classification Methods: Pattern Matching, Recurrent Neural Network, Artificial Neural Networks**

How AI Chatbot Works

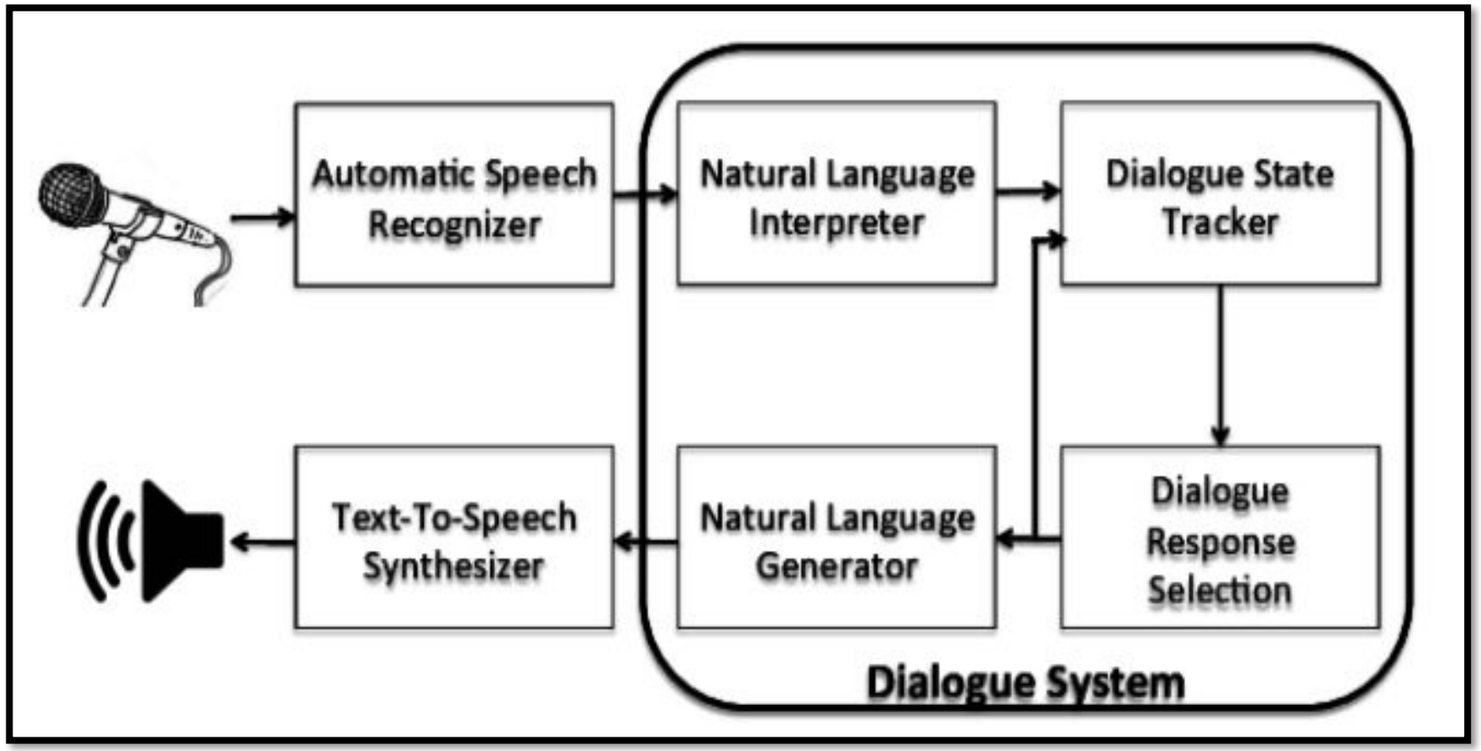


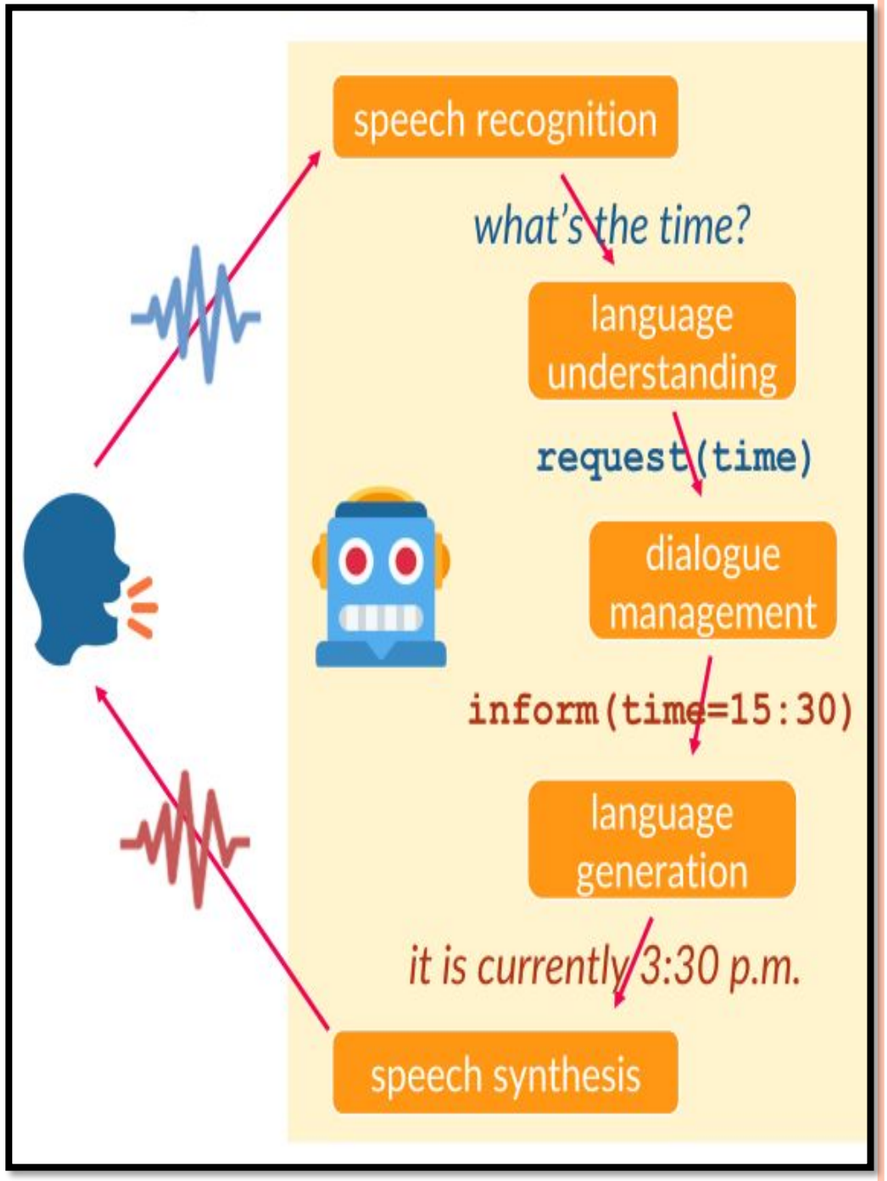


Example


Flight
arrive_city=LA
depart_flight_date=4/14
depart_city=Seattle
Hotel
hotel_city=LA
checkin_date=4/14
hotel_category= ?
Rental Car
pickup_city=LA
pickup_date=4/14
car_size= ?
Text Input
I'd like to book a flight to LA on 4/14.
Natural Language Understanding(NLU)
EOS
<slot>
O
O
<intent>
Semantic Frame
Inform(arrive_city=LA, depart_flight_date=4/14)
Natural Language Generation(NLG)
EOS
User Goal
User Dialogue Action
Inform(depart_city=Seattle)
System Action/Policy
request_depart_city
Dialogue Management (DM)
User Agenda Modeling
User Simulator
Dialogue System

CONTD..


Automatic Speech Recognizer
Natural Language Interpreter
Dialogue State Tracker
Text-To-Speech Synthesizer
Natural Language Generator
Dialogue Response Selection
Dialogue System

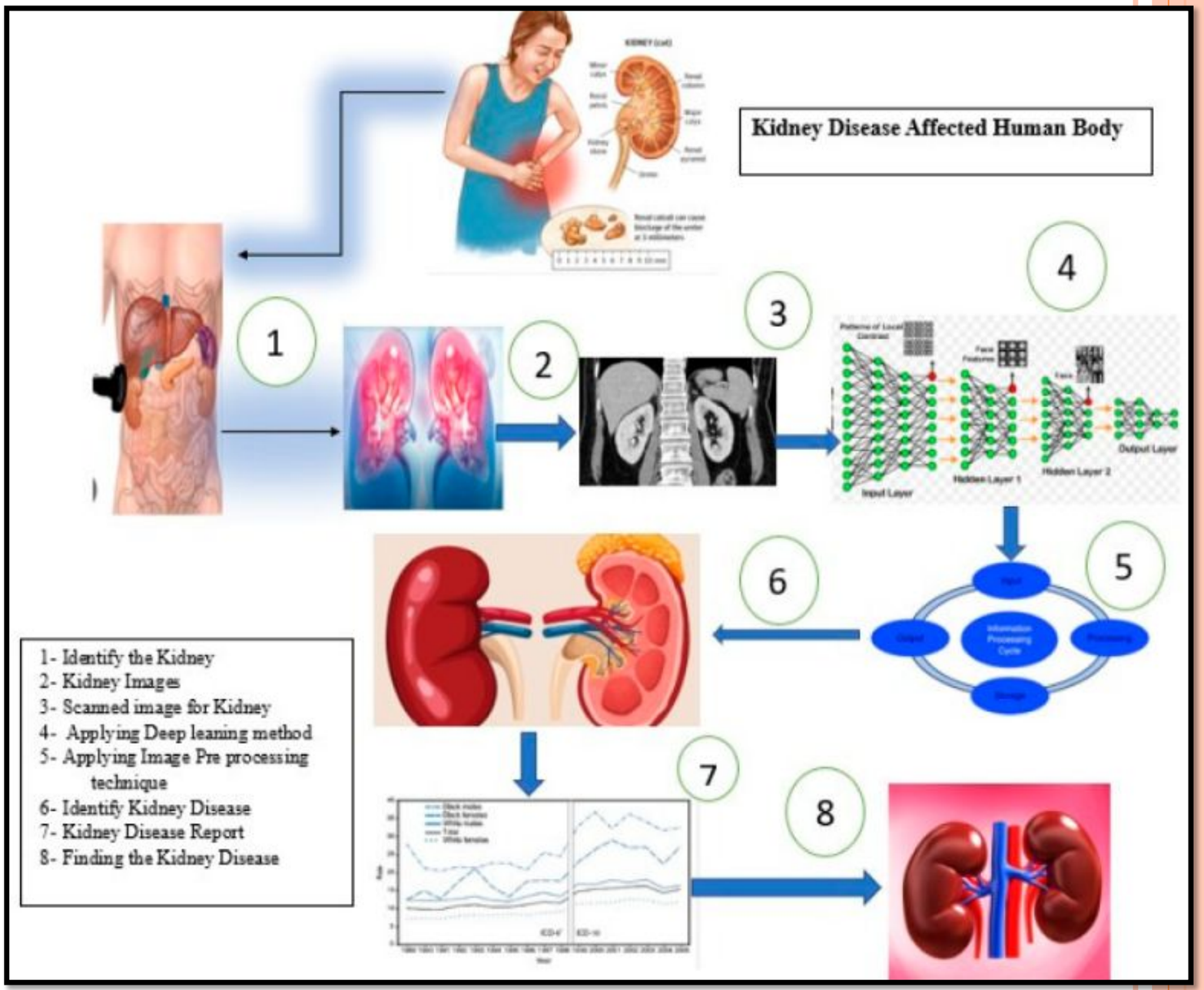

speech recognition
what's the time?
language understanding
request(time)
dialogue management
inform(time=15:30)
language generation
it is currently 3:30 p.m.
speech synthesis

4. Medical Care:

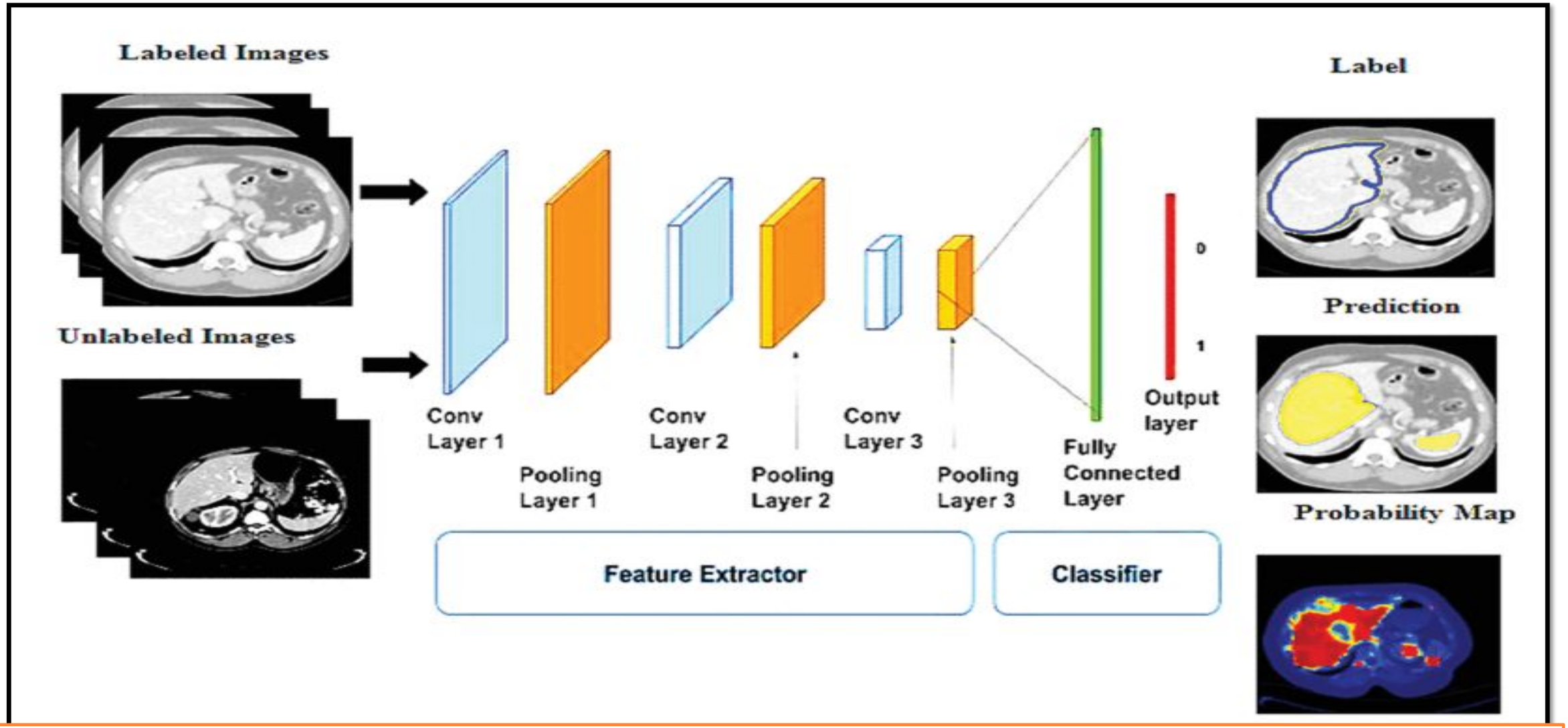
a) Deep learning helps in **detecting cancer cells** and analyzing the **MRI images to give elaborative results**.

b) Google has made **Google AI eye doctor software**. It examines **retina scans and identifies diabetic retinopathy**, which can **cause blindness**.

Suppose we have **taken Kidney Disease Prediction it will follow the structure.**



CONTD..



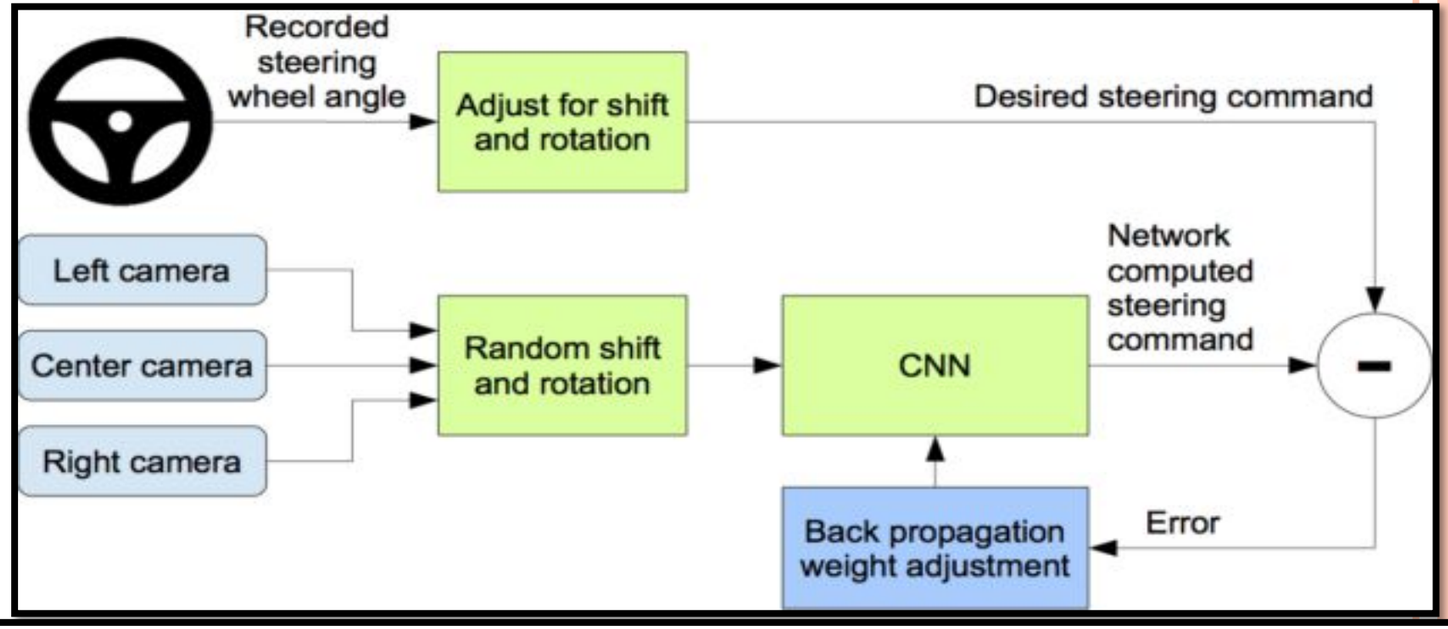
- **In this image ,** we can see that **Convolutional Neural Networks** are used for **Feature Extraction and Classification.**

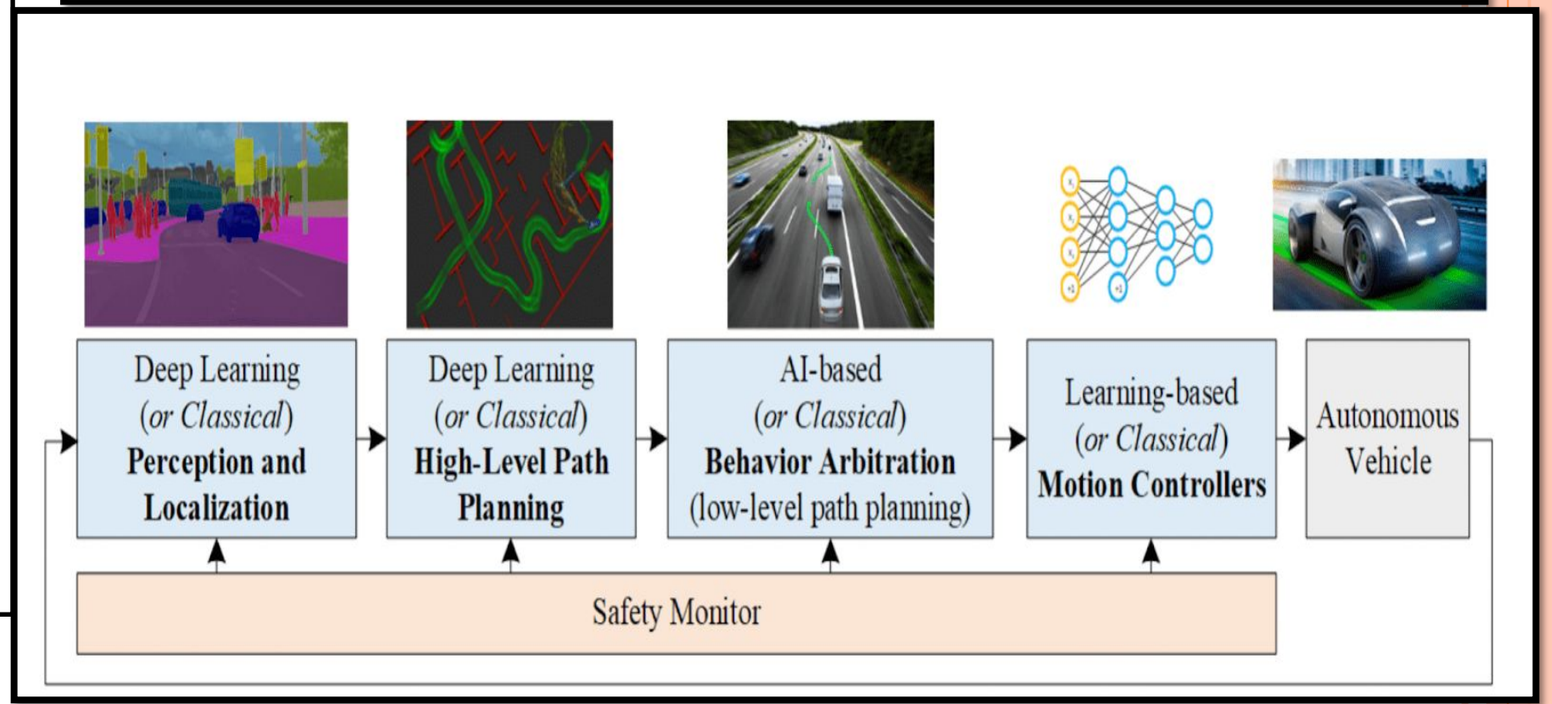
5. Self Driving Cars:

Google has made an **amazing self-driven car**. This car operates on a **combination of sensors and software**.

Car can classify objects, people, traffic signs and signals. It also detects the road works. It uses **Lidar Technology**.

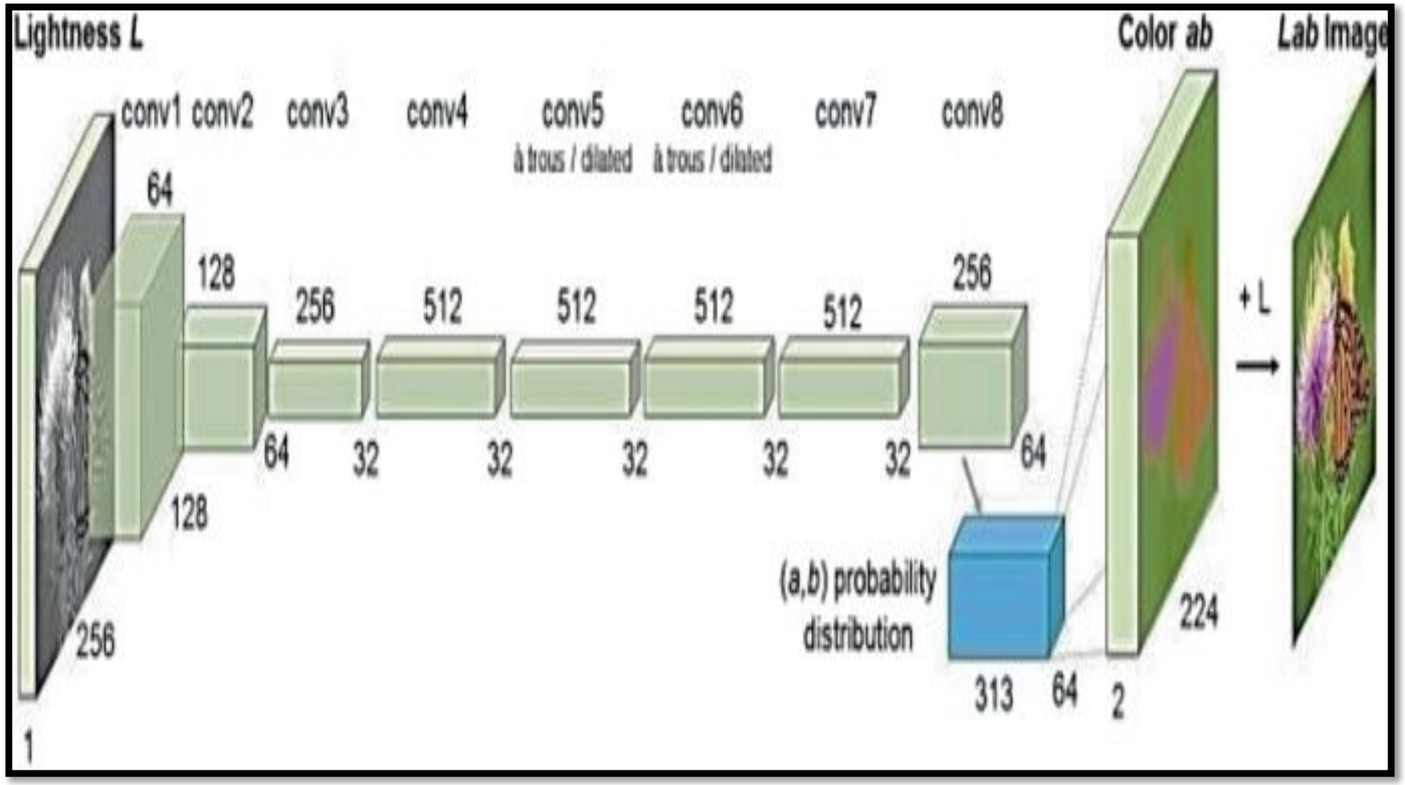
- A self-driving car **(sometimes called an autonomous car or driverless car)**.
- **Tesla**, the most **popular car manufacturing company** is **working on self-driving car**.





6. Colorizing the Images

Deep learning makes it **possible to color the black and white images.** It uses **convolutional neural network** for the same.





7. Read Lip Movements:

Lip Reading is **decoding text from the movement of the speaker's mouth.** It plays a crucial role in **human communication and speech understanding**.

Using **deep learning** with **python, oxford and Google's scientists** developed **a neural network** known as **lipnet**. It can read **people's lips with 93% success.**

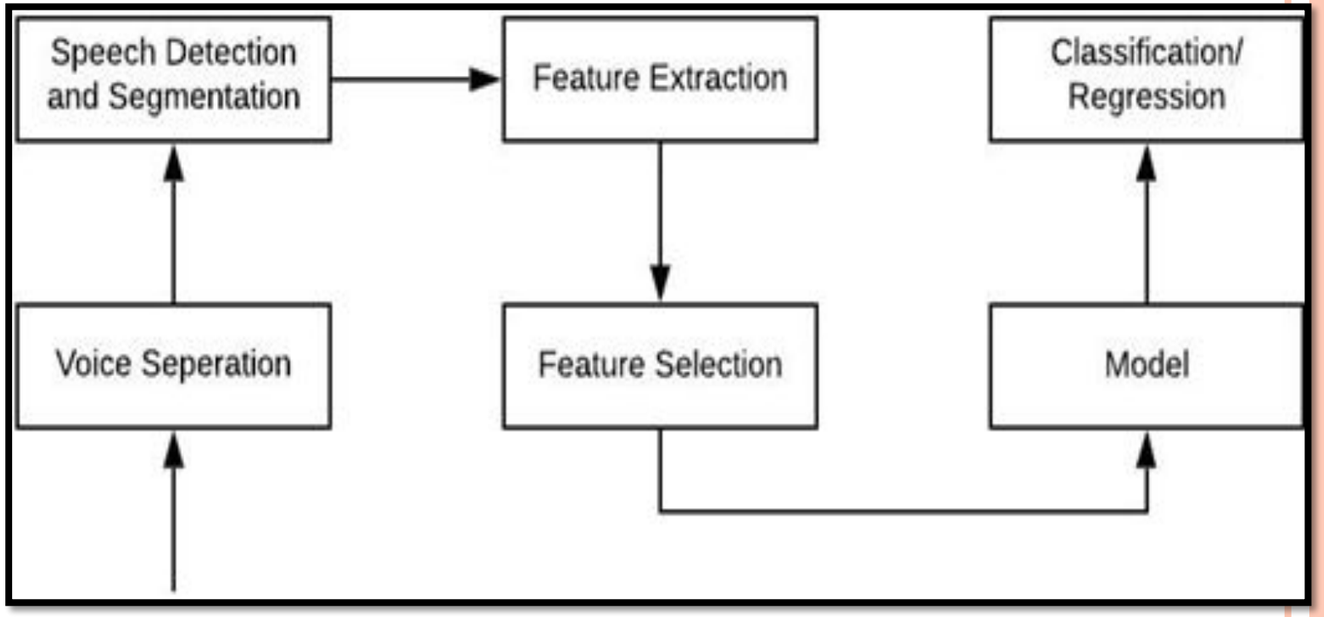
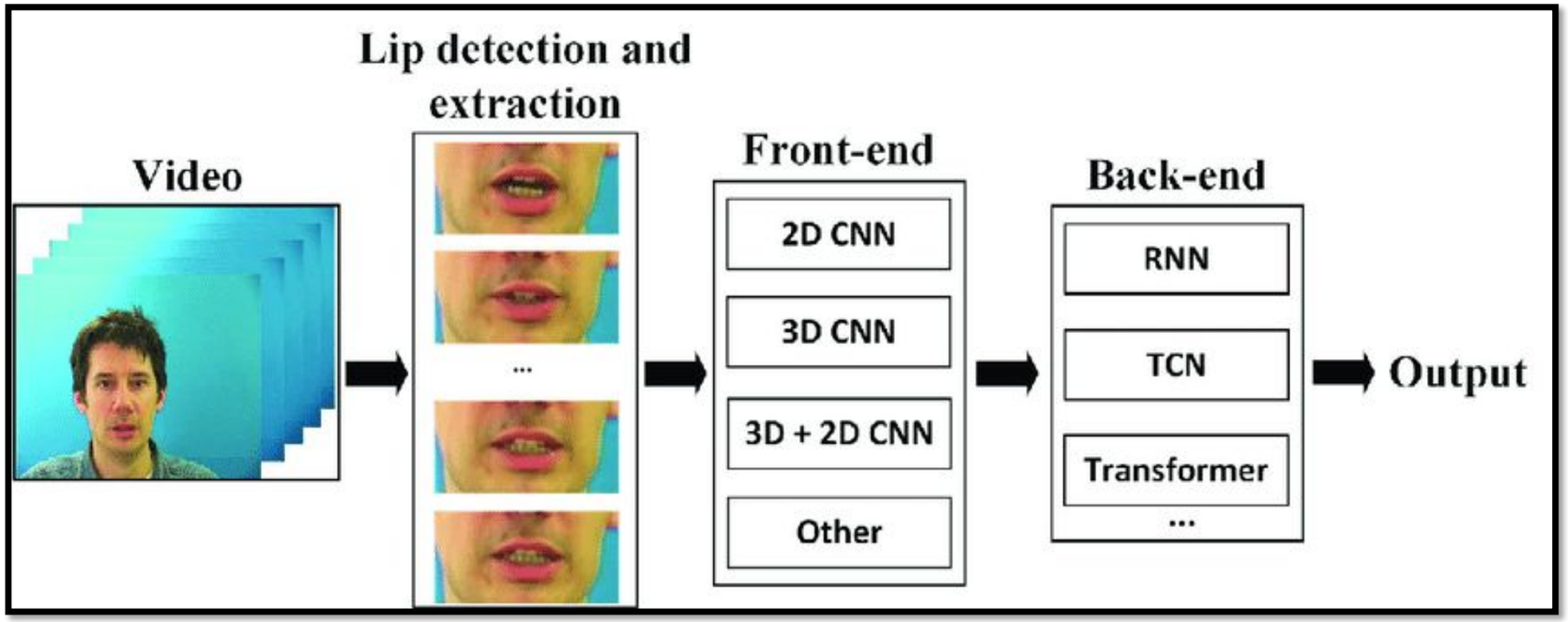
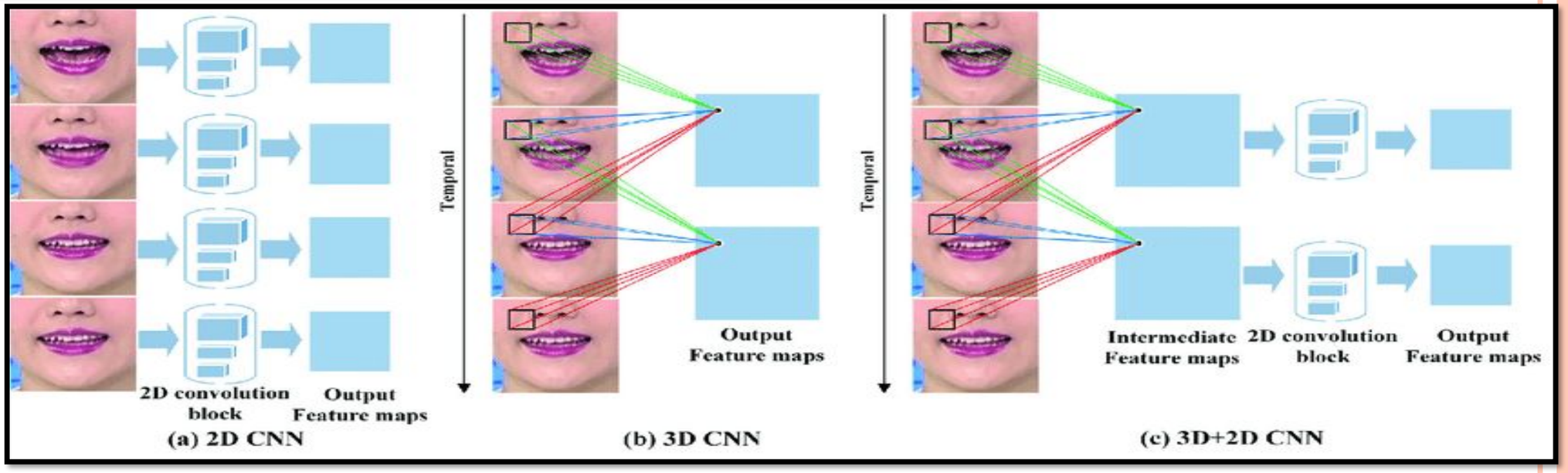


Fig: Lip Reading Process

This can be done by using Convolution Neural Networks, Long Short Term Memory, Auto Encoders, Recurrent Neural networks etc.

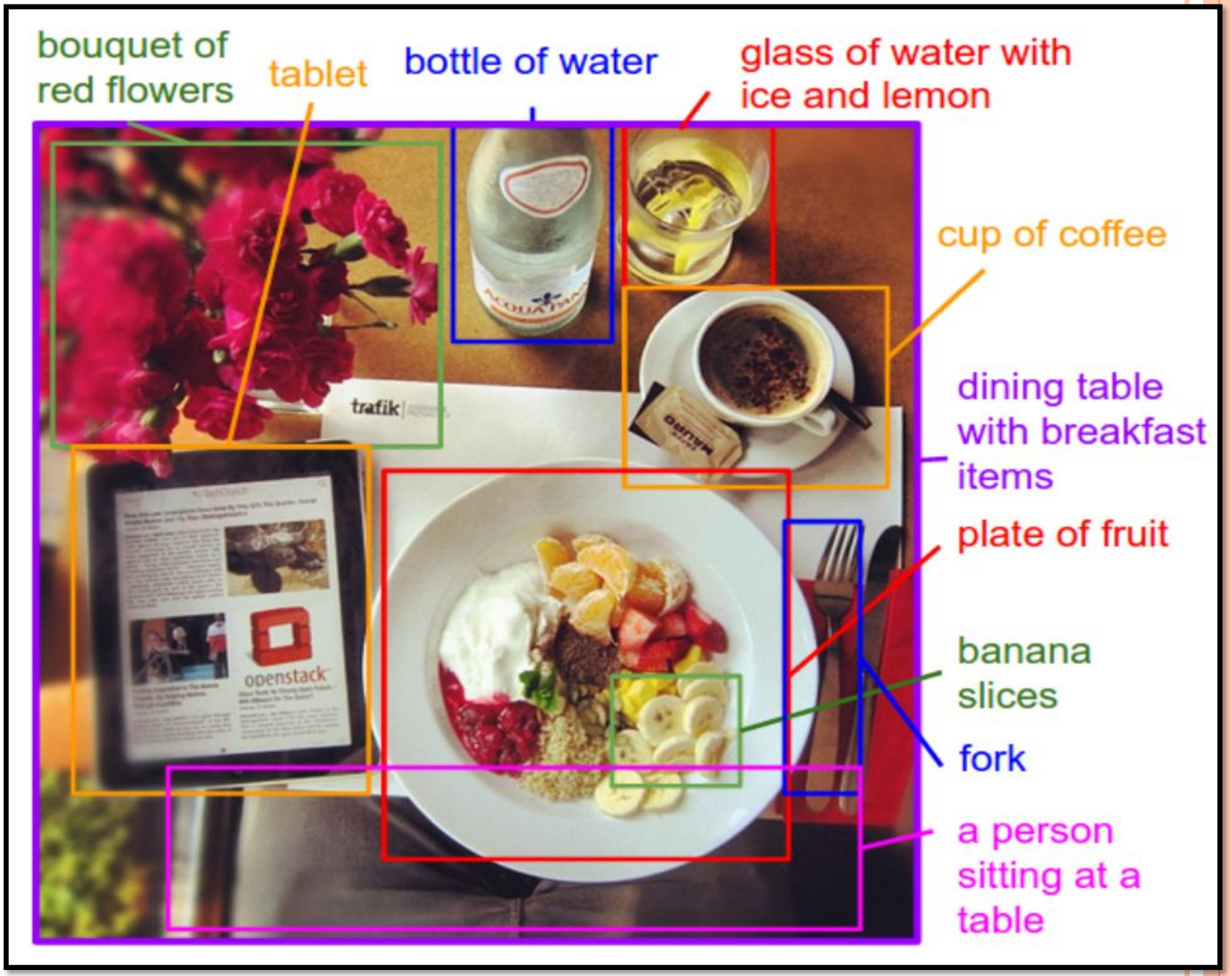
CONTD..


Lip detection and extraction
Video
Front-end
2D CNN
3D CNN
3D + 2D CNN
Other
Back-end
RNN
TCN
Transformer
...
Output


2D convolution block
Output Feature maps
(a) 2D CNN
Temporal
Output Feature maps
(b) 3D CNN
Temporal
Intermediate Feature maps
2D convolution block
Output Feature maps
(c) 3D+2D CNN

8. Photo Descriptions:

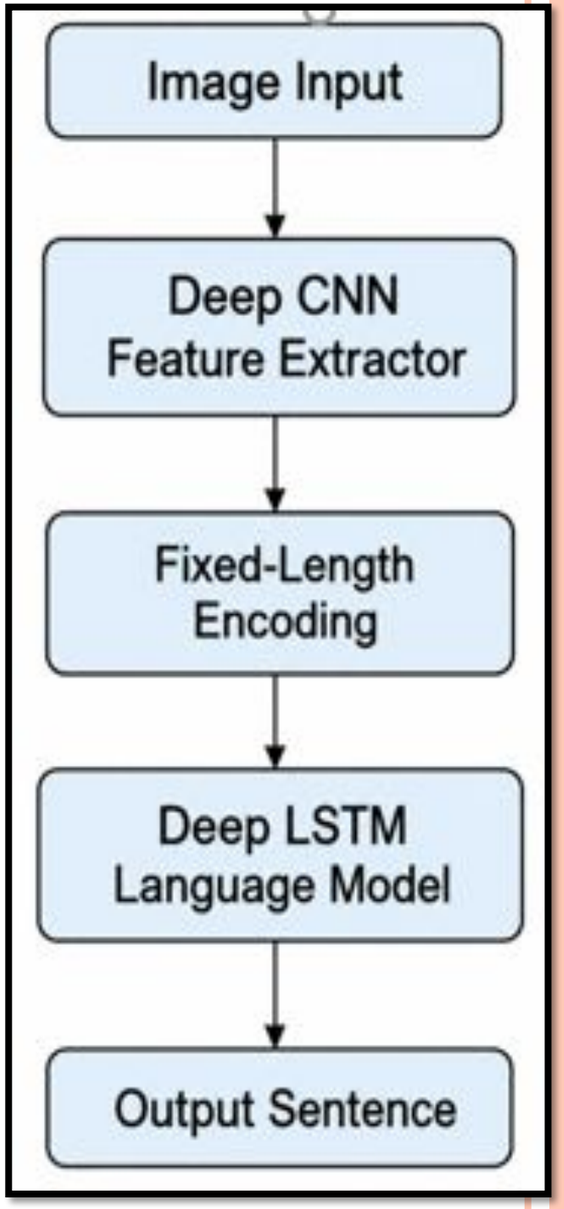
System tends to **automatically classify photographs**. Deep Learning has the **capacity to narrate every existing element in the image. Deep learning networks** can **identify the captivating areas** of the **images and can describe them into the sentences.**



CONTD..

Here we can use **Neural Captioning Model**. In this Model again **divided into 2 types**

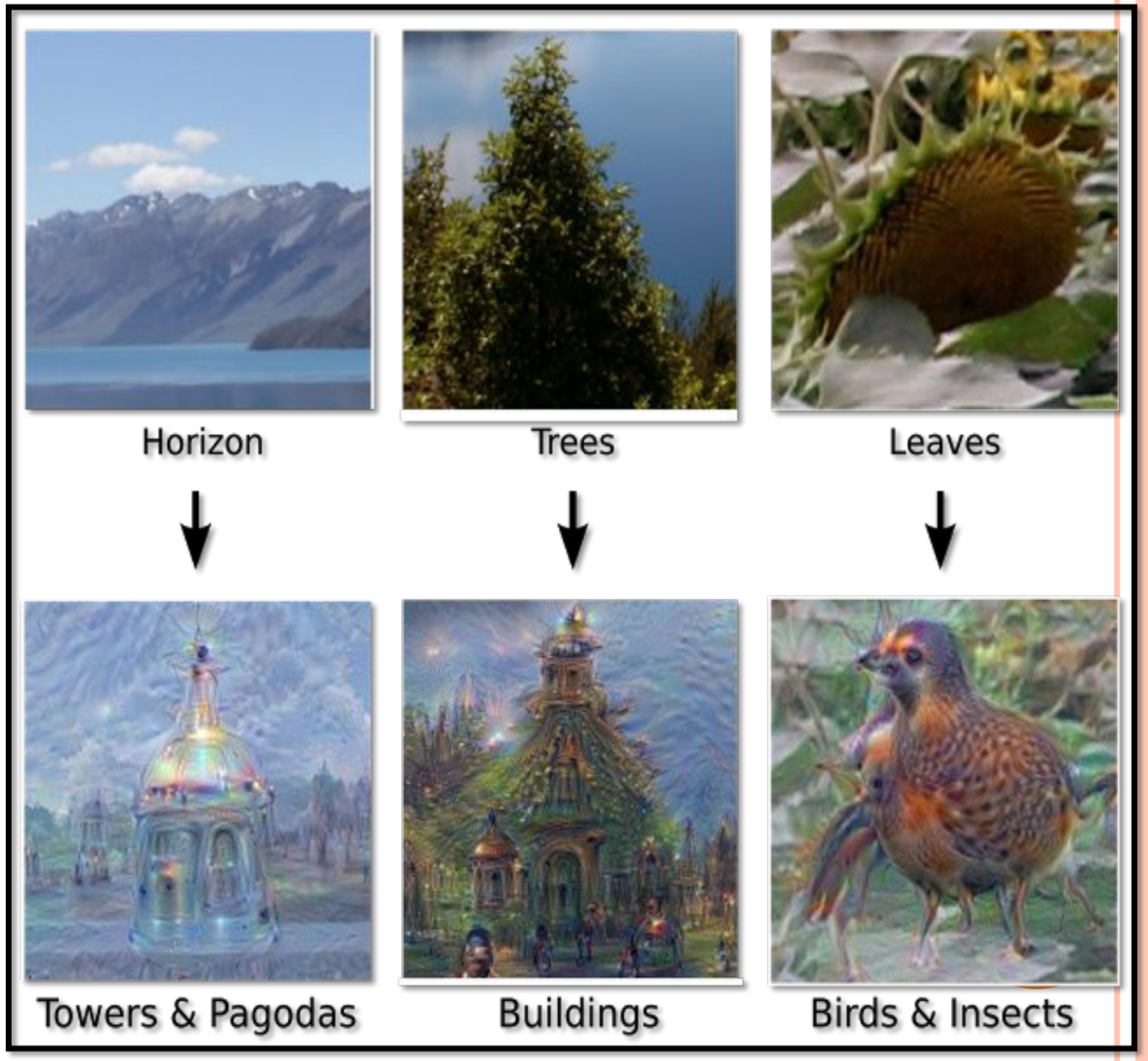
- **Feature Extraction:** The feature extraction model is a neural network that **given an image is able to extract the salient features**, often in the form of a **fixed-length vector.**
- **Language Model:** a language model predicts **the probability of the next word in the sequence given the words already present in the sequence**. It is popular to use a **recurrent neural network,** such as **a Long Short-Term Memory network, or LSTM**, as the language model. Each **output time step generates a new word in the sequence.**



9. Deep Dreaming:

It is a **very interesting application of deep learning**. As the name suggests, it **allows the system hallucinates on the top of an image and generate the resembling dream**. Dreams depend **upon the type of neural network** .

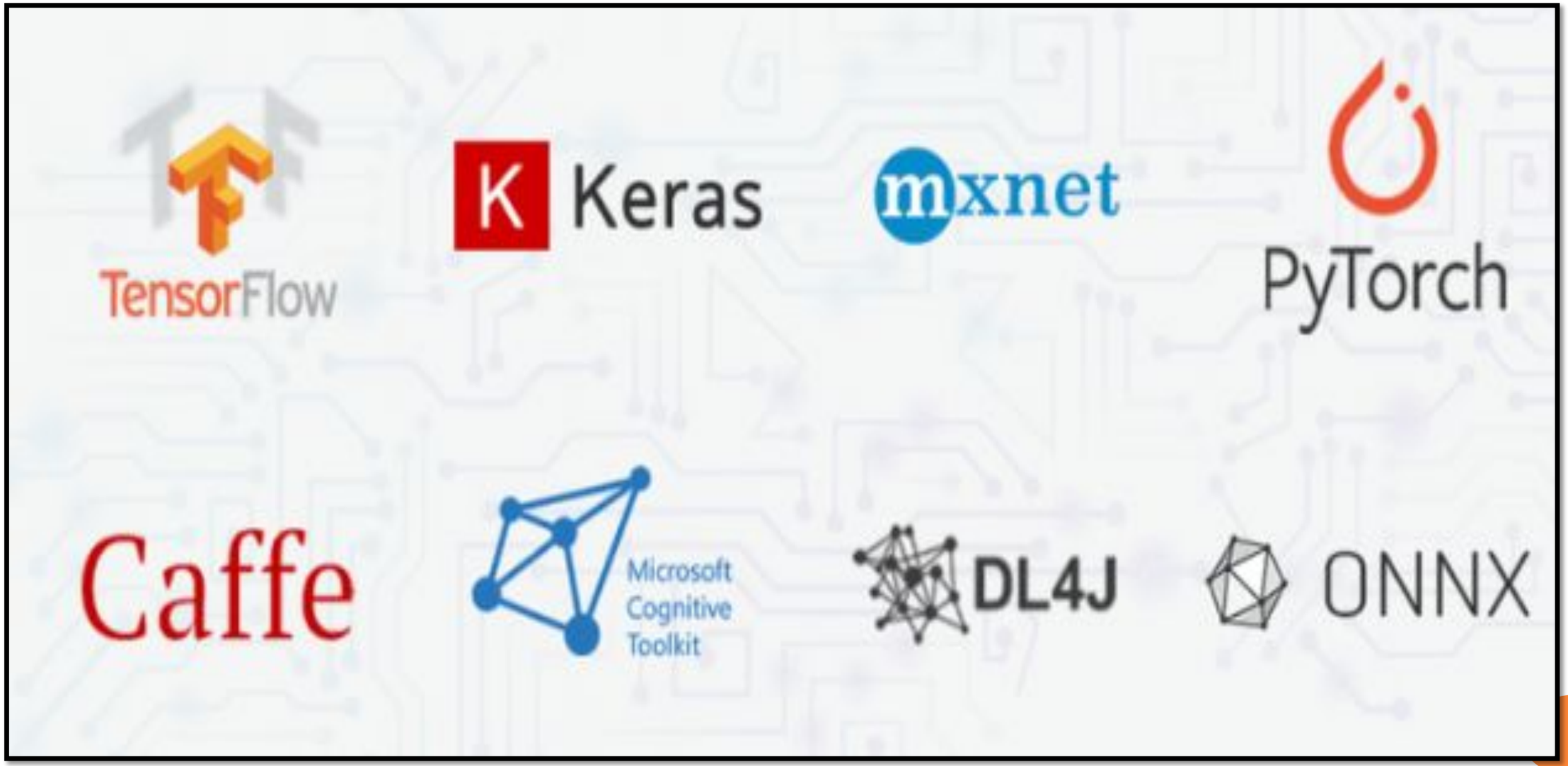
For example, **Horizon lines** tend to **get filled with towers and pagodas**. **Rocks and trees** turn into **buildings**. **Birds and insects** appear in **images of leaves**.



10. Advertising:

- Deep learning has **transformed the advertising field**. **Publishers and advertisers** use deep learning to **increase the relevancy of their ads.**
- Deep learning makes it possible for publishers to leverage the **content to create the real time bidding for the ads.**

Deep Learning Frame works



CONTD..

1. Tensor Flow: It is one of the most popular **deep learning frameworks**. Developed by **the Google Brain team,** TensorFlow supports languages **such as Python, C++, and R** to create **deep learning models along with wrapper libraries**. It is available on both **desktop and mobile.**

The most well-known **use case of Tensor Flow** has got to be **Google Translate** coupled with capabilities such as **natural language processing, text classification, summarization, speech/image/handwriting recognition, forecasting, and tagging.**



2. TORCH/PyTorch:



- PyTorch is a **Python-based scientific computing package** serving **two broad purposes:**
- A replacement for **NumPy** to use the power of GPUs and other accelerators. It is implemented by using **python**.
- An **automatic differentiation library** that is useful to **implement neural networks.**
- It is a **Lua based deep learning framework** and is used widely amongst **industry giants such as Facebook, Twitter, and Google.**
- **It is widely used Natural Language Processing, Computer Vision Applications**

CONTD..

3. DEEPLEARNING4J:

DEEPLEARNING4J

- The **j in Deeplearning4j** stands for **Java.** Needless to say, it is a **deep learning library for the Java Virtual Machine (JVM).** It is developed in **Java and supports other JVM languages like Scala, Clojure, and Kotlin.**
- **Deeplearning4j** comes with **deep network** support through **Convolution Neural Networks (CNN), Recurrent Neural Networks (RNN), Recursive Neural Tensor Network (RNTN) and Long Short-Term**

4. The Microsoft Cognitive Toolkit/ Cntk:

- Popularly known for easy **training and a combination of popular model types across servers,** the **Microsoft Cognitive Toolkit (earlier known as CNTK)** is an **open-source deep learning framework to train deep learning models.**
- It performs **efficient Convolution Neural Networks** and **training for image, speech, and text-based data.**
- **It** Supported by interfaces such as **Python, C++.**

5.Keras:

- **Keras library** was developed, and Written in **Python**, the **Keras neural networks library** supports **both convolutional and recurrent networks** that are capable of running on **either TensorFlow or Theano.**
- The **primary usage of Keras** is in **classification, text generation, and summarization, tagging, translation along with speech recognition, and others.**
- **The main used of Keras is ,** it supports **multiple deep learning backends**

6.ONNX:

- **ONNX or the Open Neural Network Exchange** was developed as an **open-source deep learning ecosystem**. Developed by **Microsoft and Facebook,** ONNX proves to be a **deep learning framework** that enables **developers to switch easily between platforms.**
- ONNX models are **natively supported in The Microsoft Cognitive Toolkit, Caffe2, MXNet, and PyTorch**. It also **provides converters** for **different machine learning frameworks** like **TensorFlow, CoreML, Keras, and Scikit Learn.**

ONNX

CONTD..

7.mxnet:

- **MXNet (pronounced as mix-net)** is a **deep learning framework** that is supported by **Python, R, C++, and Julia.**
- MXNet supports **Long Short-Term Memory (LTSM) networks,** along with both **RNN and CNN**. This deep learning framework is known for its capabilities in **imaging, handwriting/speech recognition, forecasting as well as NLP.**

8.Caffee:

- **Caffe (Convolutional Architecture for Fast Feature Embedding)** is an open source deep learning framework that supports a variety of **deep learning architectures such as CNN, RCNN, LSTM** and **fully connected networks.** Caffe is most popular for **image classification and segmentation tasks.**
- **Caffe** is primarily **a C++ library** and Therefore, **Caffe offers interfaces** for daily use by way of the **command line, Python and MATLAB.**

CONTD..

Many big businesses across a wide range of industries that use Machine Learning are, in fact, powered by Deep Learning, such as:

- ✔ **Tesla**, which powers its **autopilot system in self-driving cars;**
- ✔ **Apple** — with its **face recognition, starting from iOS 10**;
- ✔ **Google**, **Apple**, and **Amazon** for their **virtual voice assistants**;
- ✔ **Netflix** - for their advanced **recommendation system**;
- ✔ **JP Morgan Chase** — for insider **trading detection and government regulatory compliance**;
- ✔ **Pinterest** — for discovering **visually similar objects, colors, patterns, and more**;
- ✔ **Baker Hughes** — for **seismic modeling, automated well planning, predicting machinery failure, optimizing supply chains**, and many more;
- ✔ **Via** — uses Deep Learning **to optimize ride sharing routes**.

Learning beyond Curriculum

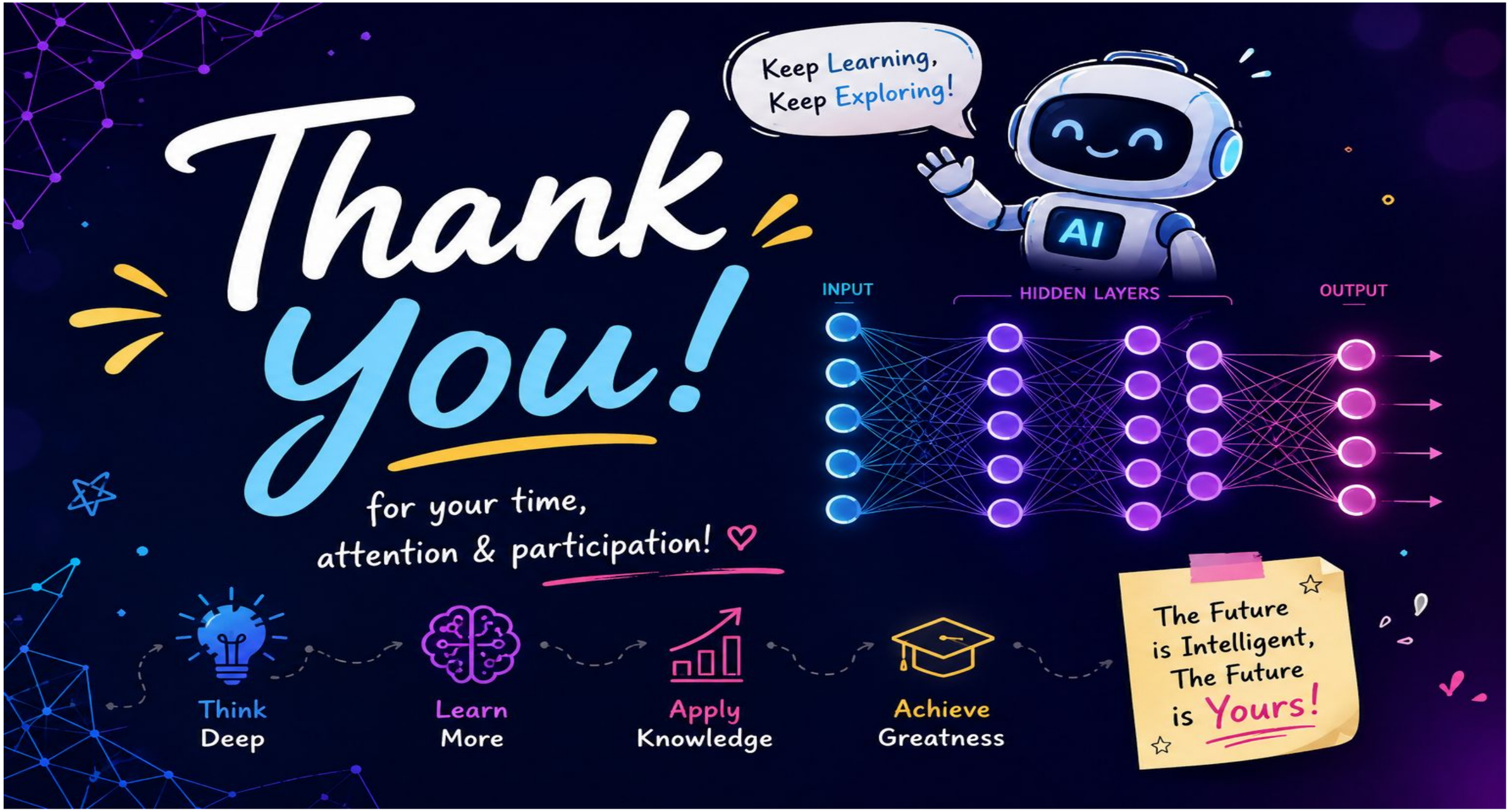
1. Coursera
2. NPTEL
3. MIT Online Certification

Courses

1. Introduction to Deep Learning
2. Deep Learning specialization
3. Neural Networks and Deep Learning
4. TensorFlow developer professional Certificate
5. Deep Learning for Object Detection

1. Introduction to Deep Learning - IIT Ropar
2. Deep Learning for Computer Vision
3. Deep Learning – IIT Kharagpur

1. Deep Learning for AI and Computer Vision
2. Introduction to Deep Learning.
3. Deep Learning – Mastering Neural Networks.
4. Advances in Imaging and Deep Learning: Medical, VR-AR, and Self-Driving Cars


Keep Learning,
Keep Exploring!
AI
Thank You!
for your time,
attention & participation!
INPUT
HIDDEN LAYERS
OUTPUT
Think
Deep
Learn
More
Apply
Knowledge
Achieve
Greatness
The Future
is Intelligent,
The Future
is Yours!